Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Práctica 2: Caos en sistemas biológicos

Información general

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Asignatura: **Gemelos Digitales**

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Descripción

Comparar métodos de diferencias finitas para solucionar sistemas de ODEs no lineales de primer orden que exhiben diferentes conjuntos compactos invariantes para diferentes valores en sus parámetros y/o condiciones iniciales. El modelo matemático está compuesto por las siguientes tres EDOs de primer orden:

$$\dot{x} = r_1 x(1 - b_1 x) - a_{12} xy - a_{13} xz$$

$$\dot{y} = r_2 y(1 - b_2 y) - a_{21} xy$$

$$\dot{z} = (r_3 - a_{31}) xz - d_3 z + \rho_i$$

donde la variable de estado $x(t)$ representa la población de células anormales o patalógicas, la variable de estado $y(t)$ la población de células normales o sanas y la variable de estado $z(t)$ población de células efectoras (citotóxicas o NK); la descripción y valores de los parámetros se presenta a continuación:

$r_1 = 1$: Tasa de crecimiento de las células patológicas.

$b_1 = 1$: Capacidad de carga máxima de las células patológicas.

$a_{12} \in [0, 1]$: Fracción de células patológicas eliminadas por las células sanas.

$a_{13} = 2.5$: Fracción de células patológicas eliminadas por células efectoras.

$r_2 = 0.6$: Tasa de crecimiento de las células sanas.

$b_2 = 1$: Capacidad de carga máxima de las células sanas.

$a_{21} = 1.5$: Fracción de células sanas eliminadas por las células patológicas.

$r_3 = 3$: Tasa de reclutamiento máximo de células efectoras.

$a_{31} = 0.3$: Fracción de células efectoras inactivadas por las células patológicas.

$d_3 = 0.4$: Tasa de muerte de las células efectoras.

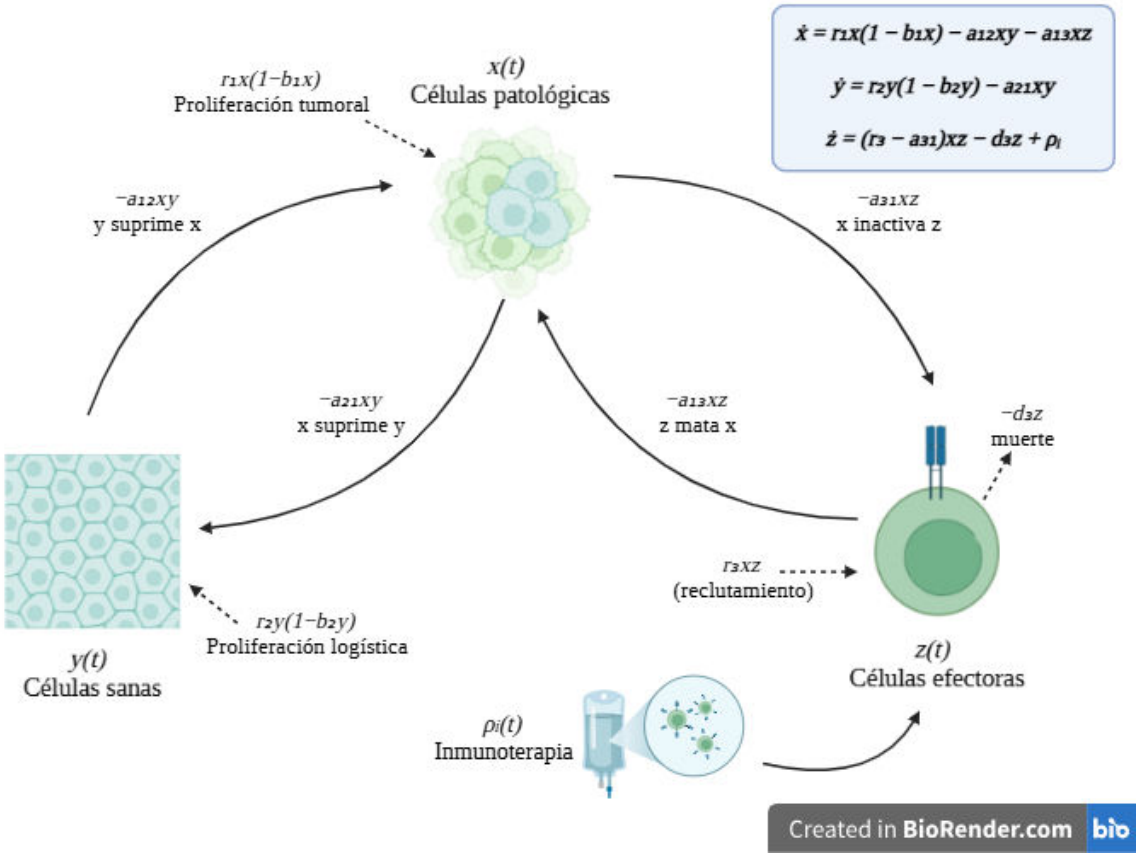
$\rho_i(t)$: Tratamiento de inmunoterapia.

Cabe destacar que la dinámica del sistema se localiza en octante no negativo definido por el siguiente dominio:

$$\mathbf{R}^3_{+,0} = \{x(t) \geq 0, y(t) \geq 0, z(t) \geq 0\}$$

El sistema exhibe diferentes dinámicas para distintos valores del parámetro de bifurcación a_{12}, como se indica a continuación:

1. Un atractor caótico con $a_{12} = 1$, en el intervalo de tiempo $t \in [0, 1000]$ y las condiciones iniciales: $(x(0), y(0), z(0)) = (0.555, 0.148, 0.1'72)$.
2. Una órbita periódica con $a_{12} = 0.9$, en el intervalo de tiempo $t \in [0, 500]$ y las condiciones iniciales: $(x(0), y(0), z(0)) = (0.254781, 0.115167, 0.600527)$.
3. Un ciclo límite interno con $a_{12} = 0.8$, en el intervalo de tiempo $t \in [0, 500]$ y las condiciones iniciales: $(x(0), y(0), z(0)) = (0.2, 0.6, 0.1)$.
4. Un ciclo límite externo con $a_{12} = 0.75$, en el intervalo de tiempo $t \in [0, 500]$ y las condiciones iniciales: $(x(0), y(0), z(0)) = (0.4, 0.2, 0.4)$.
5. Convergencia hacia un punto de equilibrio con $a_{12} = 0.5$, en el intervalo de tiempo $t \in [0, 500]$ y las condiciones iniciales: $(x(0), y(0), z(0)) = (0.5, 0.5, 0.4)$.



[Figura representativa del sistema elaborada en BioRender]

Figura. Generar la figura del sistema presa-depredador con https://bioart.niaid.nih.gov/ o https://www.biorender.com/.

Referencias

[1] Valle, Paul A., *Syllabus para Biología de Sistemas y Gemelos Digitales*, Tecnológico Nacional de México / Instituto Tecnológico de Tijuana, Tijuana, B.C., México, 2026. Permalink: https://biomath.xyz/course/

[2] Itik, M., & Banks, S. P. (2010). Chaos in a three-dimensional cancer model. *international Journal of bifurcation and chaos*, *20*(01), 71-79. https://doi.org/10.1142/S0218127410025417

[3] Bryan, Kurt. *Differential equations: A toolbox for modeling the world*. Simiode, 2022. Permalink: https://www.simiode.org/resources/8307


```
clc; clear; close all; warning('off')
```

Puntos de equilibrio

```
syms x y z
syms r1 b1 a12 a13
syms r2 b2 a21
syms r3 a31 d3 rhoi
dx_dt = r1*x*(1-b1*x) - a12*x*y - a13*x*z == 0;
dy_dt = r2*y*(1-b2*y) - a21*x*y == 0;
dz_dt = (r3 - a31)*x*z - d3*z +rhoi == 0;
```

```
E = solve([dx_dt,dy_dt,dz_dt],[x,y,z]);
fprintf(['Puntos de equilibrio del sistema: ',num2str(length(E.x))]);
```

```
Puntos de equilibrio del sistema: 6
```

```
for i = 1:6
    disp([E.x(i);E.y(i);E.z(i)])
end
```

$$\begin{pmatrix} \frac{r_1 - \frac{a_{13}\,\sigma_1}{2\,(a_{13}\,a_{31} - a_{13}\,r_3)}}{b_1\,r_1} \\ 0 \\ \frac{\sigma_1}{2\,(a_{13}\,a_{31} - a_{13}\,r_3)} \end{pmatrix}$$

where

$$\sigma_1 = a_{31}\,r_1 - r_1\,r_3 - \sqrt{r_1\,\left(r_1\,{a_{31}}^2 + 2\,r_1\,a_{31}\,b_1\,d_3 - 4\,a_{13}\,\mathrm{rhoi}\,a_{31}\,b_1 - 2\,r_1\,a_{31}\,r_3 + r_1\,{b_1}^2\,{d_3}^2 - 2\,r_1\,b_1\,d_3\,r_3\right.}$$

$$\begin{pmatrix} \frac{r_2\,(a_{12} - b_2\,r_1)}{\sigma_2} - \frac{\sigma_1}{2\,\sigma_2\,(a_{31} - r_3)} \\ \frac{a_{21}\,r_1 - b_1\,r_1\,r_2}{\sigma_2} + \frac{a_{21}\,\sigma_1}{2\,b_2\,r_2\,\sigma_2\,(a_{31} - r_3)} \\ -\frac{\sigma_1}{2\,a_{13}\,b_2\,r_2\,(a_{31} - r_3)} \end{pmatrix}$$

where

$$\sigma_1 = \sqrt{{a_{12}}^2\,{a_{21}}^2\,{d_3}^2 + 2\,{a_{12}}^2\,a_{21}\,a_{31}\,d_3\,r_2 - 2\,{a_{12}}^2\,a_{21}\,d_3\,r_2\,r_3 + {a_{12}}^2\,{a_{31}}^2\,{r_2}^2 - 2\,{a_{12}}^2\,a_{31}\,{r_2}^2\,r_3 + {a_{12}}^2\,{r_2}^2\,{r_3}^2}$$

$$\sigma_2 = a_{12}\,a_{21} - b_1\,b_2\,r_1\,r_2$$

$$\begin{pmatrix} \frac{r_2\,(a_{12}-b_2\,r_1)}{\sigma_2}+\frac{\sigma_1}{2\,\sigma_2\,(a_{31}-r_3)} \\ \frac{a_{21}\,r_1-b_1\,r_1\,r_2}{\sigma_2}-\frac{a_{21}\,\sigma_1}{2\,b_2\,r_2\,\sigma_2\,(a_{31}-r_3)} \\ \frac{\sigma_1}{2\,a_{13}\,b_2\,r_2\,(a_{31}-r_3)} \end{pmatrix}$$

where

$$\sigma_1 = \sqrt{a_{12}^2\,a_{21}^2\,d_3^2+2\,a_{12}^2\,a_{21}\,a_{31}\,d_3\,r_2-2\,a_{12}^2\,a_{21}\,d_3\,r_2\,r_3+a_{12}^2\,a_{31}^2\,r_2^2-2\,a_{12}^2\,a_{31}\,r_2^2\,r_3+a_{12}^2\,r_2^2\,r_3^2}$$

$$\sigma_2 = a_{12}\,a_{21}-b_1\,b_2\,r_1\,r_2$$

$$\begin{pmatrix} 0 \\ \frac{1}{b_2} \\ \frac{\mathrm{rhoi}}{d_3} \end{pmatrix}$$

$$\begin{pmatrix} \frac{r_1-\frac{a_{13}\,\sigma_1}{2\,(a_{13}\,a_{31}-a_{13}\,r_3)}}{b_1\,r_1} \\ 0 \\ \frac{\sigma_1}{2\,(a_{13}\,a_{31}-a_{13}\,r_3)} \end{pmatrix}$$

where

$$\sigma_1 = a_{31}\,r_1-r_1\,r_3+\sqrt{r_1\,\left(r_1\,a_{31}^2+2\,r_1\,a_{31}\,b_1\,d_3-4\,a_{13}\,\mathrm{rhoi}\,a_{31}\,b_1-2\,r_1\,a_{31}\,r_3+r_1\,b_1^2\,d_3^2-2\,r_1\,b_1\,d_3\,r_3\right.}$$

$$\begin{pmatrix} 0 \\ 0 \\ \frac{\mathrm{rhoi}}{d_3} \end{pmatrix}$$

Matriz Jacobiana

```
dx_dt = r1*x*(1-b1*x) - a12*x*y - a13*x*z;
dy_dt = r2*y*(1-b2*y) - a21*x*y;
dz_dt = (r3 - a31)*x*z - d3*z +rhoi;
J = jacobian([dx_dt,dy_dt,dz_dt],[x,y,z]);
fprintf('Matriz Jacobiana: \n'); disp(J)
```

```
Matriz Jacobiana:
```

$$\begin{pmatrix} -a_{12}\,y - a_{13}\,z - r_1\,(b_1\,x - 1) - b_1\,r_1\,x & -a_{12}\,x & -a_{13}\,x \\ -a_{21}\,y & -a_{21}\,x - r_2\,(b_2\,y - 1) - b_2\,r_2\,y & 0 \\ -z\,(a_{31} - r_3) & 0 & -d_3 - x\,(a_{31} - r_3) \end{pmatrix}$$

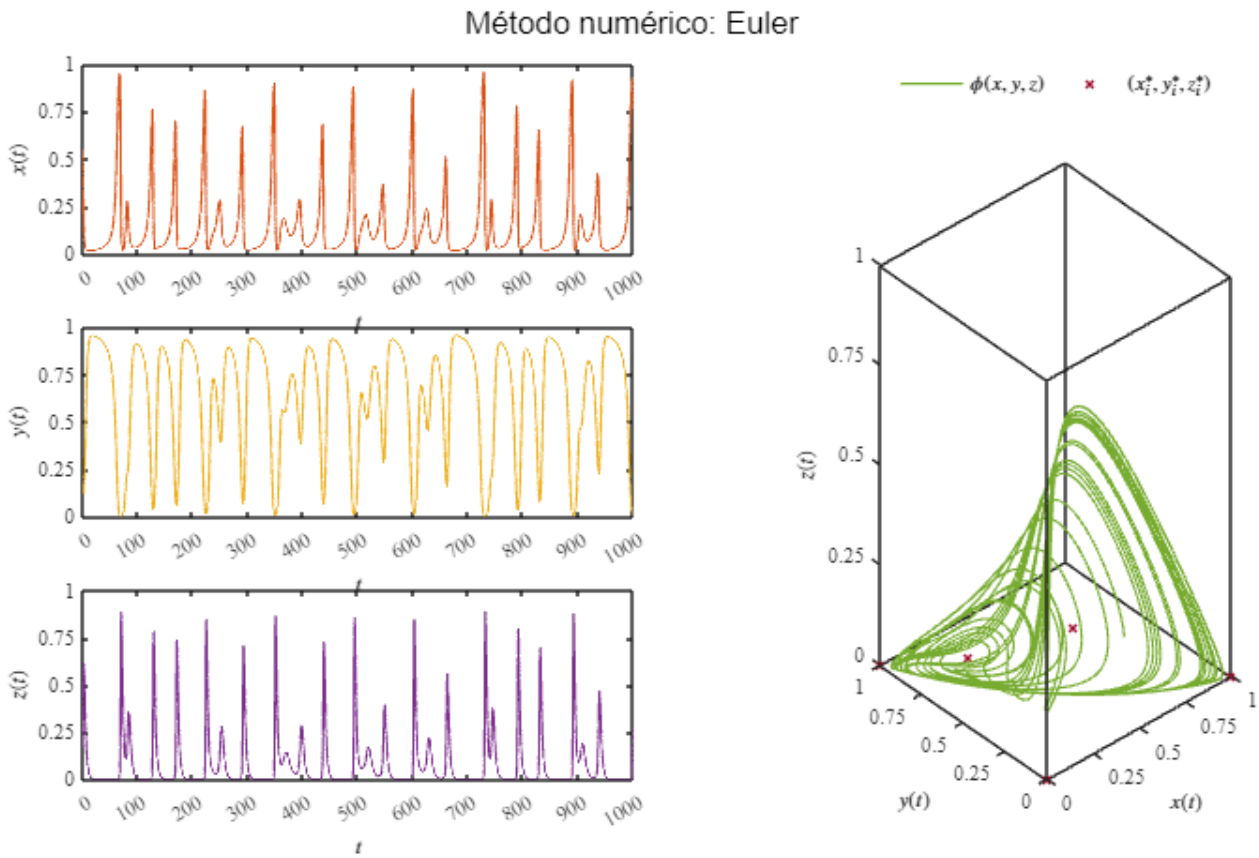
Comparación de métodos numéricos: Atractor caótico

```
D = getdata('chaos'); f0 = D.f0; par = D.par;
dt = 1e-4; tend = 1000;
rhoi = zeros(round(tend/dt)+1,1);
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-------------------------------------------------
eq1        ( 1.481e-01 , 0.000e+00 , 3.407e-01)
eq2        ( 1.481e-01 , 6.296e-01 , 8.889e-02)
eq3        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq4        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq5        ( 1.000e+00 , 0.000e+00 , 0.000e+00)
eq6        ( 0.000e+00 , 0.000e+00 , 0.000e+00)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------
eq1        -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i +3.8e-01+0.0e+00i unstable
eq2        -6.1e-01+0.0e+00i +4.1e-02+2.3e-01i +4.1e-02-2.3e-01i unstable
eq3        -6.0e-01+0.0e+00i +0.0e+00+0.0e+00i -4.0e-01+0.0e+00i unidentified
eq4        -6.0e-01+0.0e+00i +0.0e+00+0.0e+00i -4.0e-01+0.0e+00i unidentified
eq5        -1.0e+00+0.0e+00i -9.0e-01+0.0e+00i +2.3e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +6.0e-01+0.0e+00i +1.0e+00+0.0e+00i unstable
```

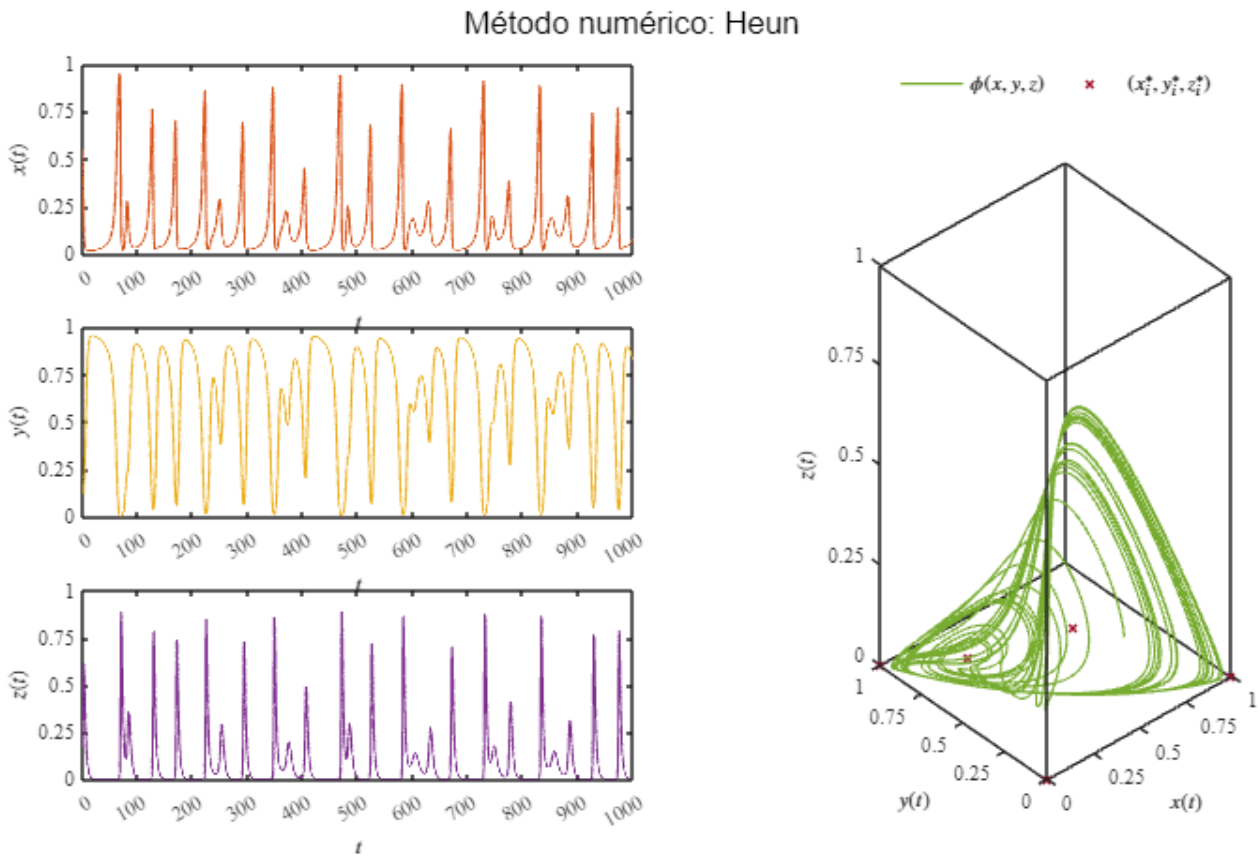
Euler

```
method = 'Euler';
[t,xe,ye,ze] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xe,ye,ze,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Chaos.pdf'],'ContentType','vector');
```



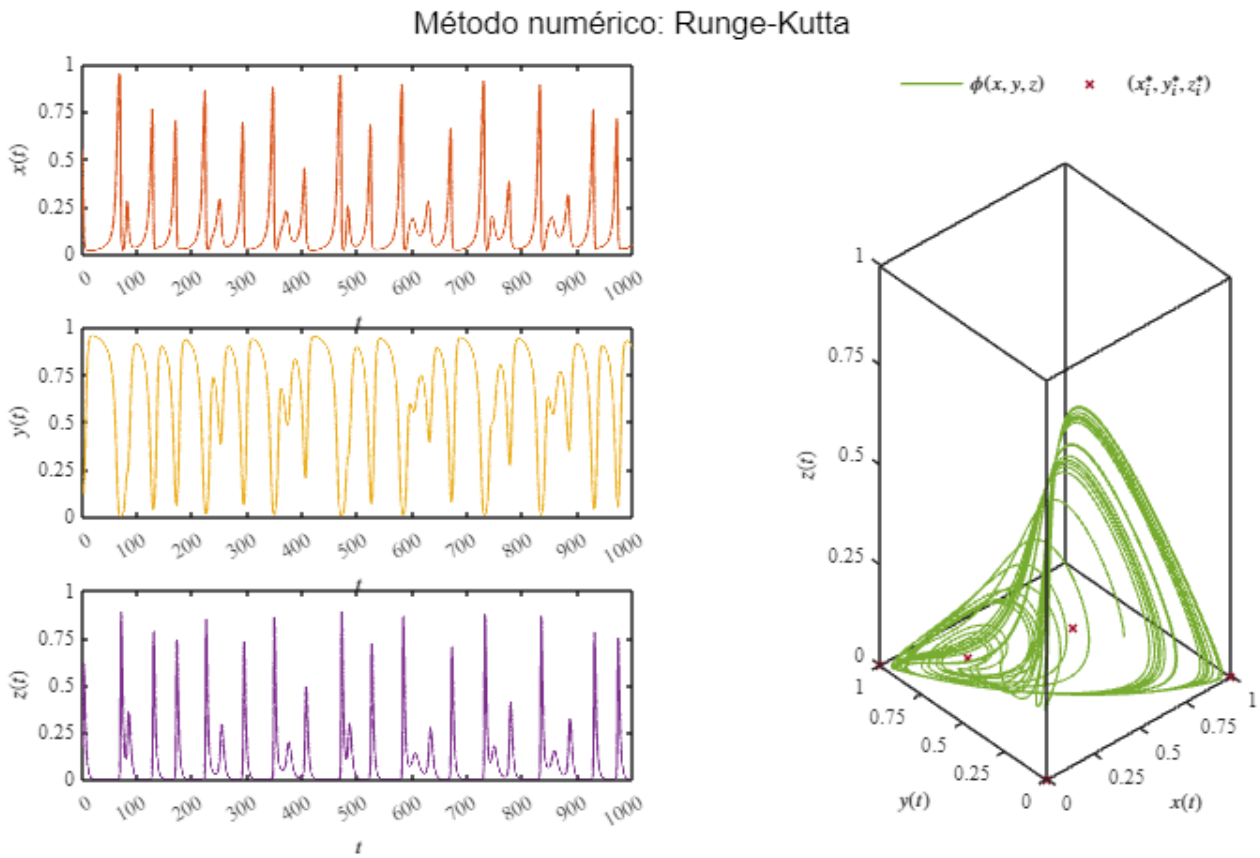
Heun

```
method = 'Heun';
[t,xh,yh,zh] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xh,yh,zh,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Chaos.pdf'],'ContentType','vector');
```



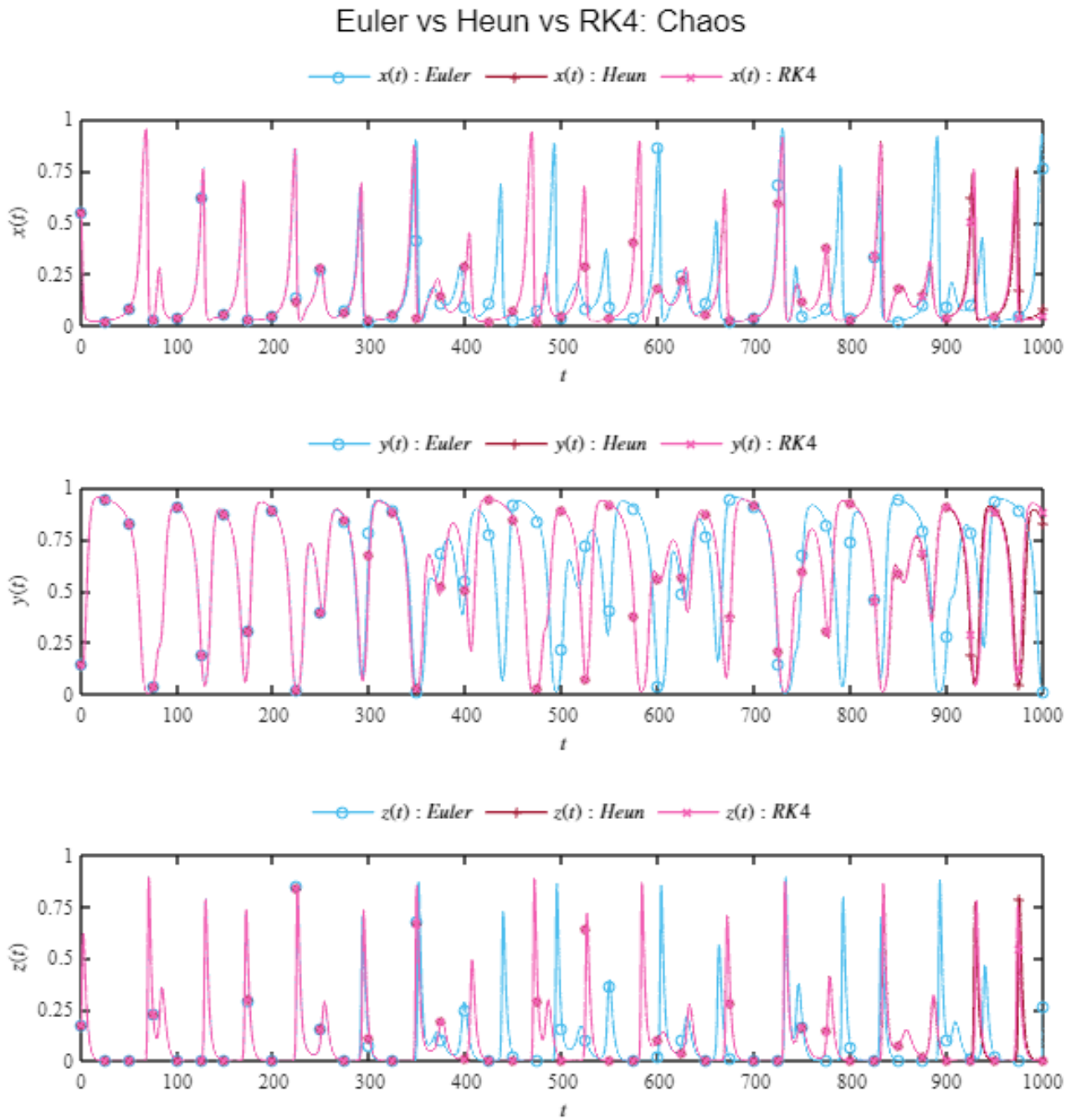
Runge-Kutta

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Chaos.pdf'],'ContentType','vector');
```



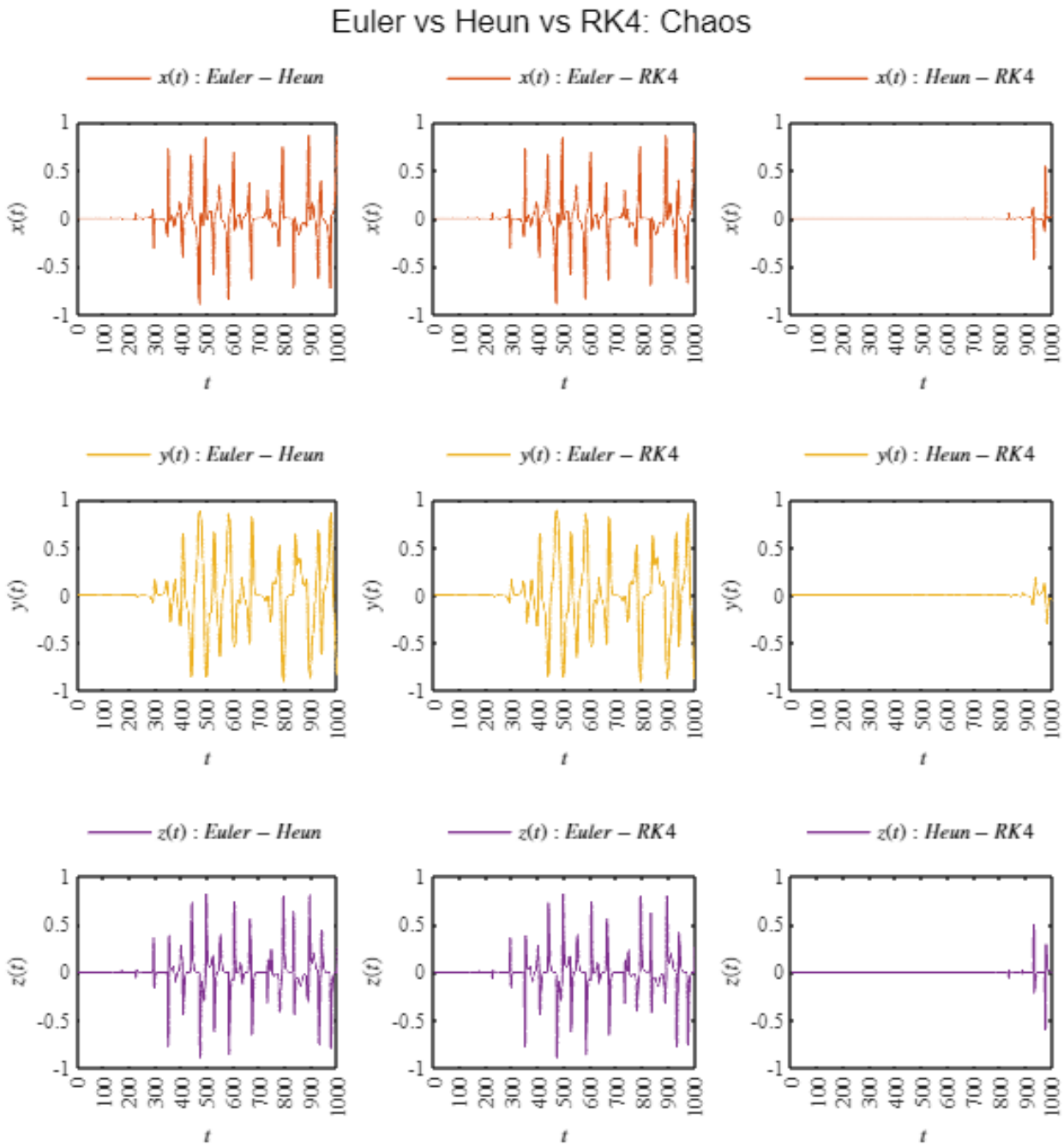
Comparación entre soluciones

```
plotdivergence(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Chaos');...
    exportgraphics(gcf,'Solutions_Chaos.pdf','ContentType','vector');
```



Error

```
ploterror(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Chaos');...
    exportgraphics(gcf,'Error_Chaos.pdf','ContentType','vector');
```



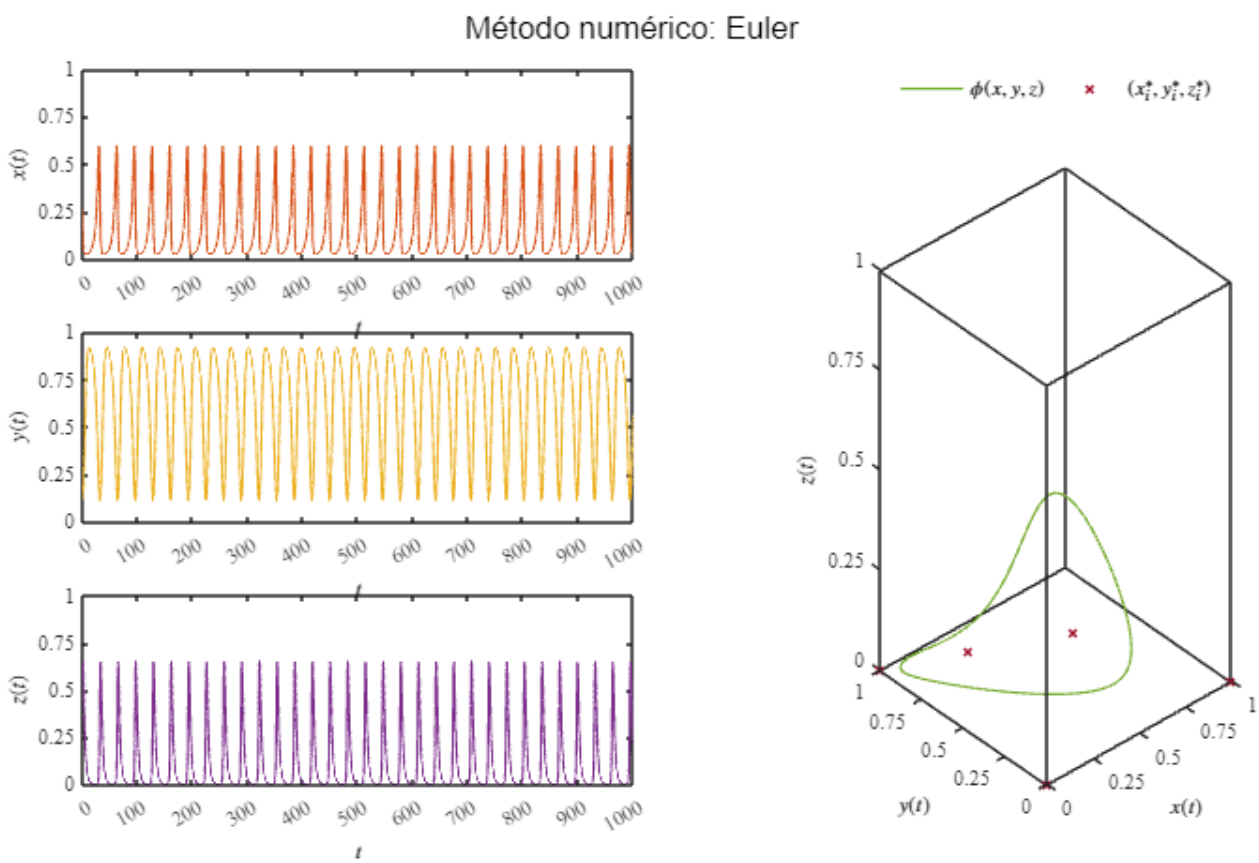
Comparación de métodos numéricos: Órbita Periódica

```
D = getdata('orbit'); f0 = D.f0; par = D.par;
dt = 1e-4; tend = 1000;
rhoi = zeros(round(tend/dt)+1,1);
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.481e-01 , 0.000e+00 , 3.407e-01)
eq2        ( 1.481e-01 , 6.296e-01 , 1.141e-01)
eq3        (-8.000e-02 , 1.200e+00 , 0.000e+00)
eq4        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq5        ( 1.000e+00 , 0.000e+00 , 0.000e+00)
eq6        ( 0.000e+00 , 0.000e+00 , 0.000e+00)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i +3.8e-01+0.0e+00i unstable
eq2        -5.8e-01+0.0e+00i +2.6e-02+2.7e-01i +2.6e-02-2.7e-01i unstable
eq3        -1.5e-01+0.0e+00i -4.9e-01+0.0e+00i -6.2e-01+0.0e+00i stable
eq4        -6.0e-01+0.0e+00i +1.0e-01+0.0e+00i -4.0e-01+0.0e+00i unstable
eq5        -1.0e+00+0.0e+00i -9.0e-01+0.0e+00i +2.3e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +6.0e-01+0.0e+00i +1.0e+00+0.0e+00i unstable
```

Euler

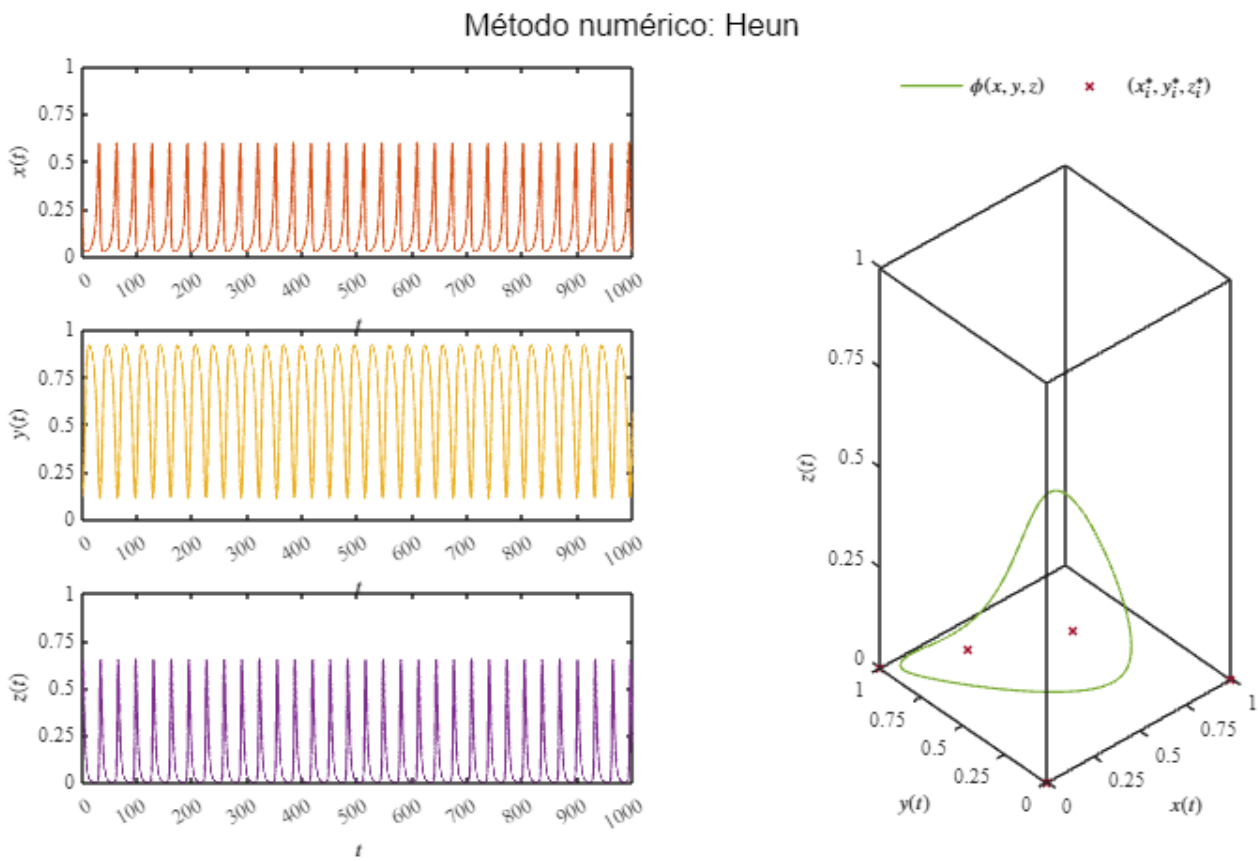
```
method = 'Euler';
[t,xe,ye,ze] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xe,ye,ze,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Orbit.pdf'],'ContentType','vector');
```



Heun

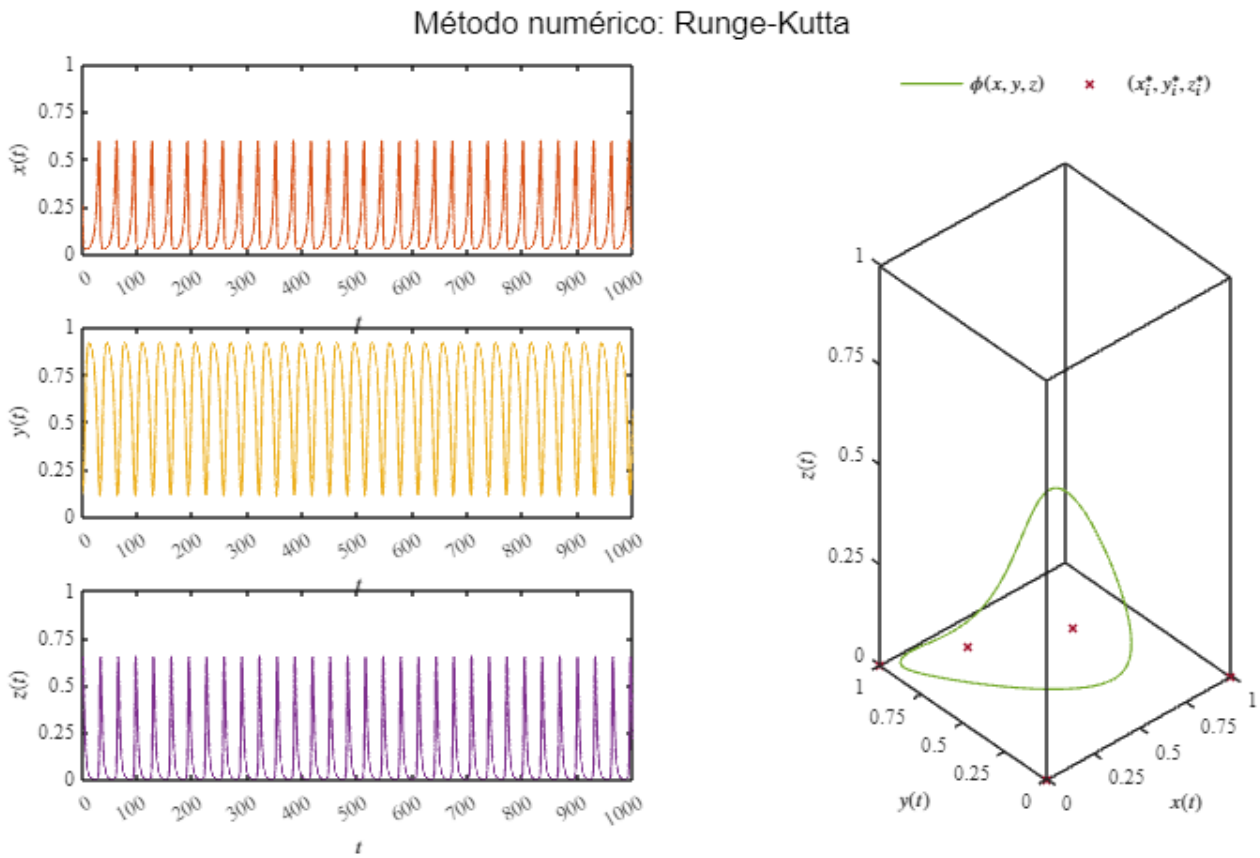
```
method = 'Heun';
[t,xh,yh,zh] = odesolver(method,f0,par,rhoi,tend,dt);
```

```
plotEDOS(t,xh,yh,zh,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Orbit.pdf'],'ContentType','vector');
```



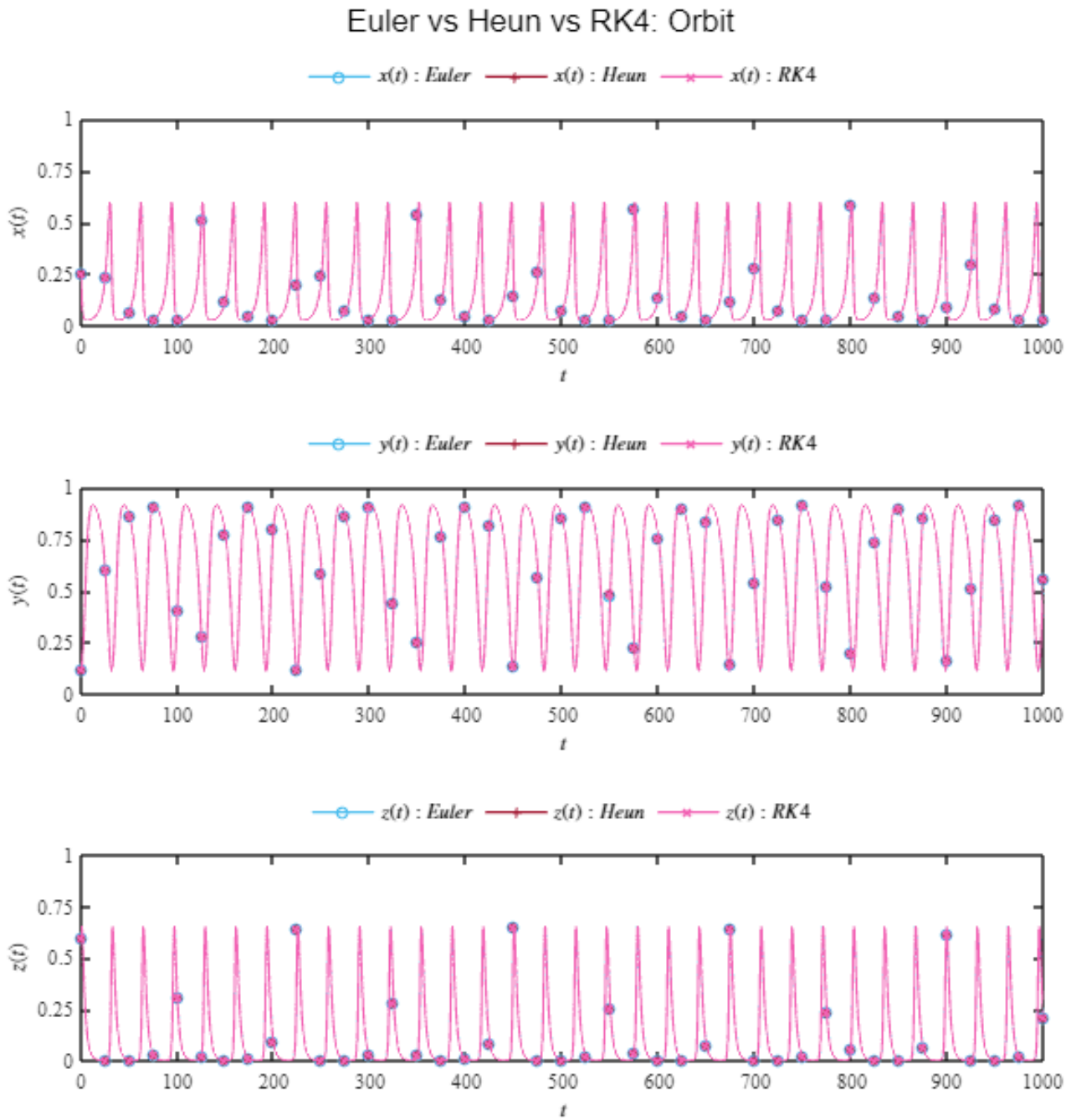
Runge-Kutta

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Orbit.pdf'],'ContentType','vector');
```



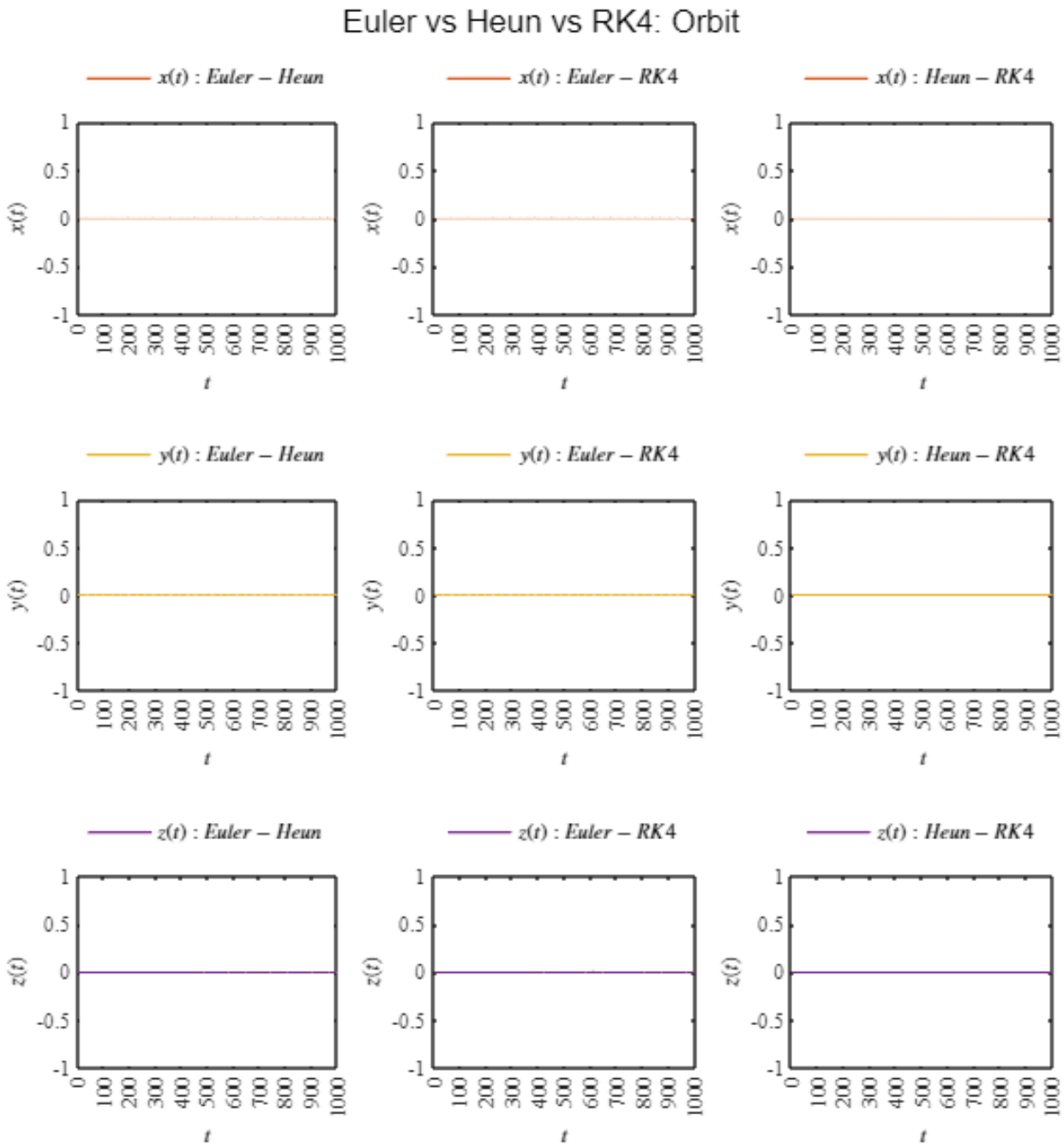
Comparación entre soluciones

```
plotdivergence(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Orbit');...
    exportgraphics(gcf,'Solutions_Orbit.pdf','ContentType','vector');
```



Error

```
ploterror(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Orbit');...
    exportgraphics(gcf,'Error_Orbit.pdf','ContentType','vector');
```



Comparación de métodos numéricos: Ciclo límite 1

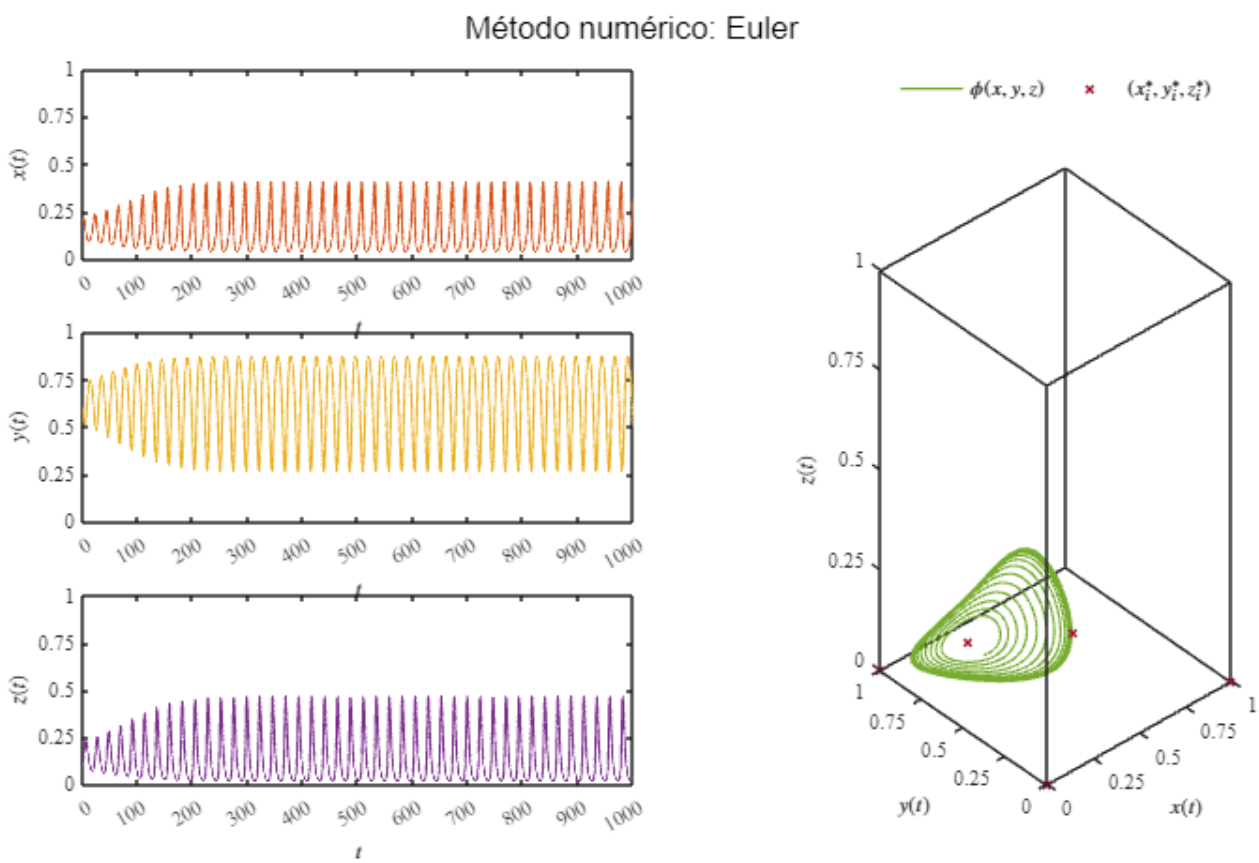
```
D = getdata('cycle1'); f0 = D.f0; par = D.par;
dt = 1e-4; tend = 1000;
rhoi = zeros(round(tend/dt)+1,1);
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.481e-01 , 0.000e+00 , 3.407e-01)
eq2        ( 1.481e-01 , 6.296e-01 , 1.393e-01)
eq3        (-2.000e-01 , 1.500e+00 , 0.000e+00)
eq4        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq5        ( 1.000e+00 , 0.000e+00 , 0.000e+00)
eq6        ( 0.000e+00 , 0.000e+00 , 0.000e+00)
```

```
Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i +3.8e-01+0.0e+00i unstable
eq2        -5.5e-01+0.0e+00i +1.1e-02+3.1e-01i +1.1e-02-3.1e-01i unstable
eq3        -3.5e-01+2.4e-01i -3.5e-01-2.4e-01i -9.4e-01+0.0e+00i stable
eq4        -6.0e-01+0.0e+00i +2.0e-01+0.0e+00i -4.0e-01+0.0e+00i unstable
eq5        -1.0e+00+0.0e+00i -9.0e-01+0.0e+00i +2.3e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +6.0e-01+0.0e+00i +1.0e+00+0.0e+00i unstable
```

Euler

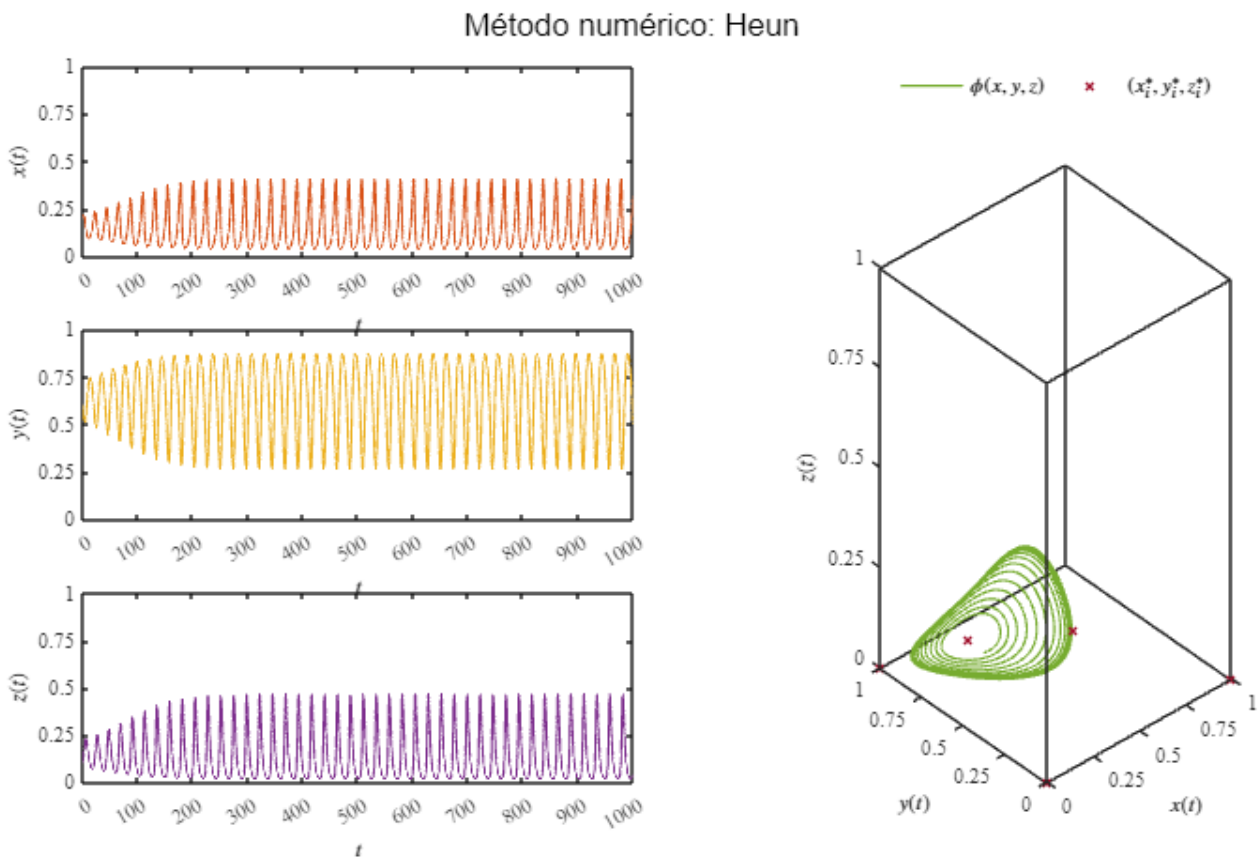
```
method = 'Euler';
[t,xe,ye,ze] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xe,ye,ze,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle1.pdf'],'ContentType','vector');
```



Heun

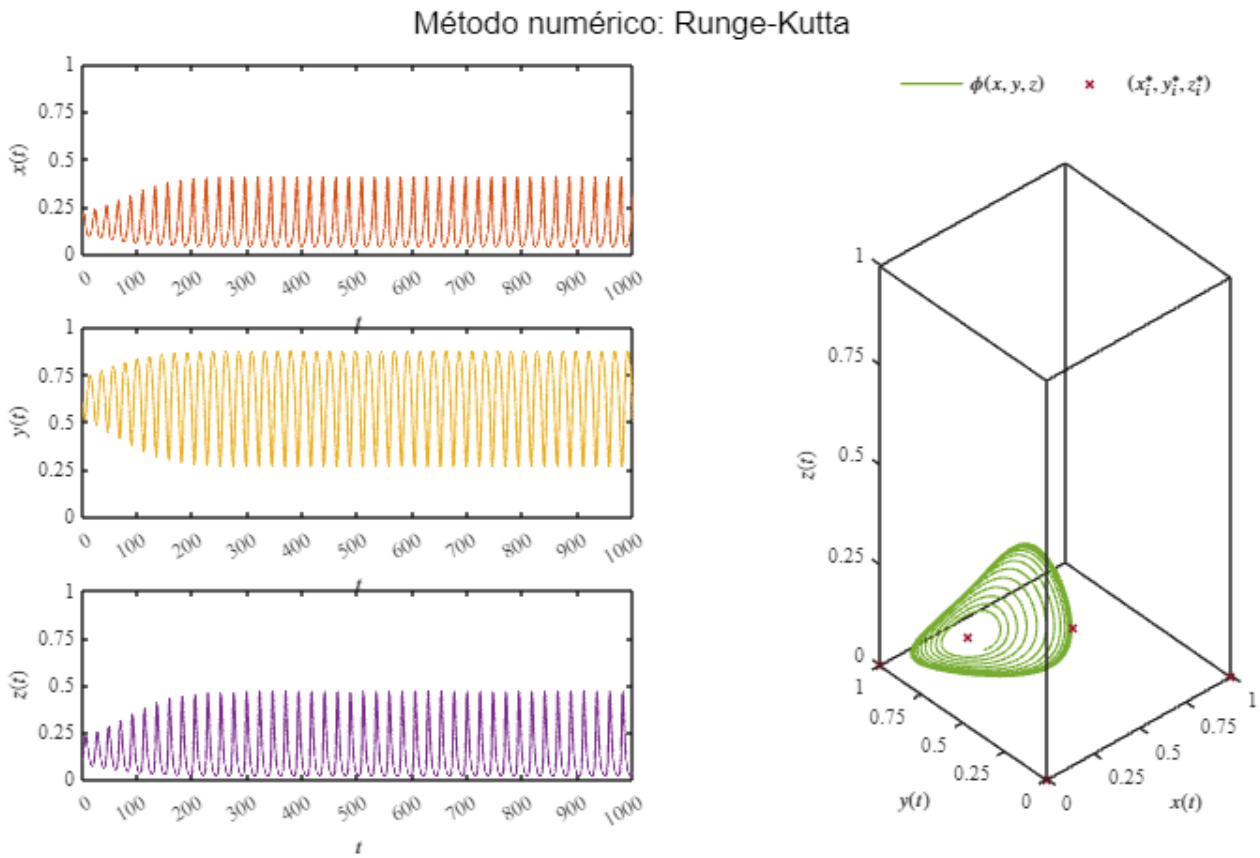
```
method = 'Heun';
[t,xh,yh,zh] = odesolver(method,f0,par,rhoi,tend,dt);
```

```
plotEDOS(t,xh,yh,zh,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle1.pdf'],'ContentType','vector');
```



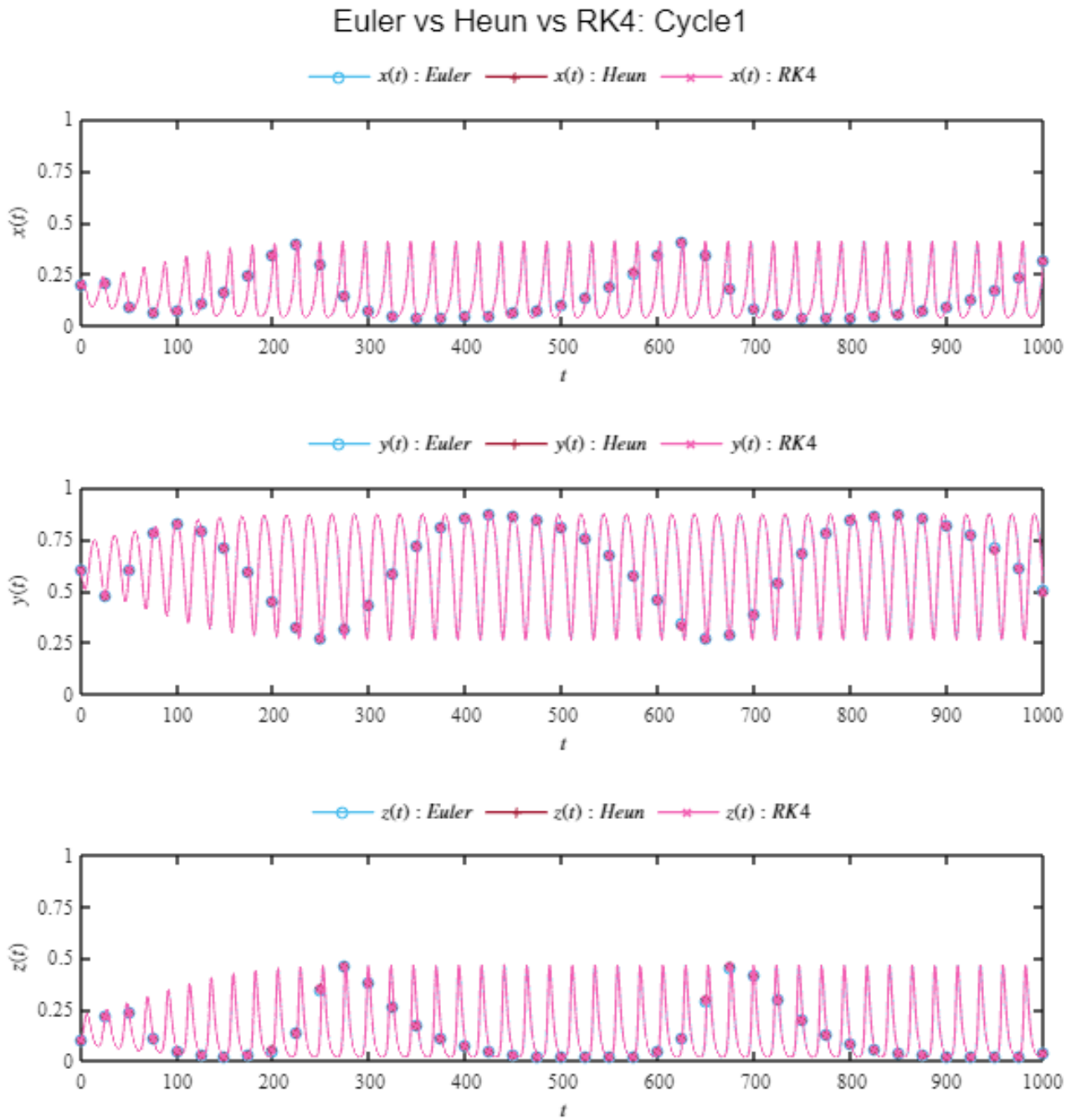
Runge-Kutta

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle1.pdf'],'ContentType','vector');
```



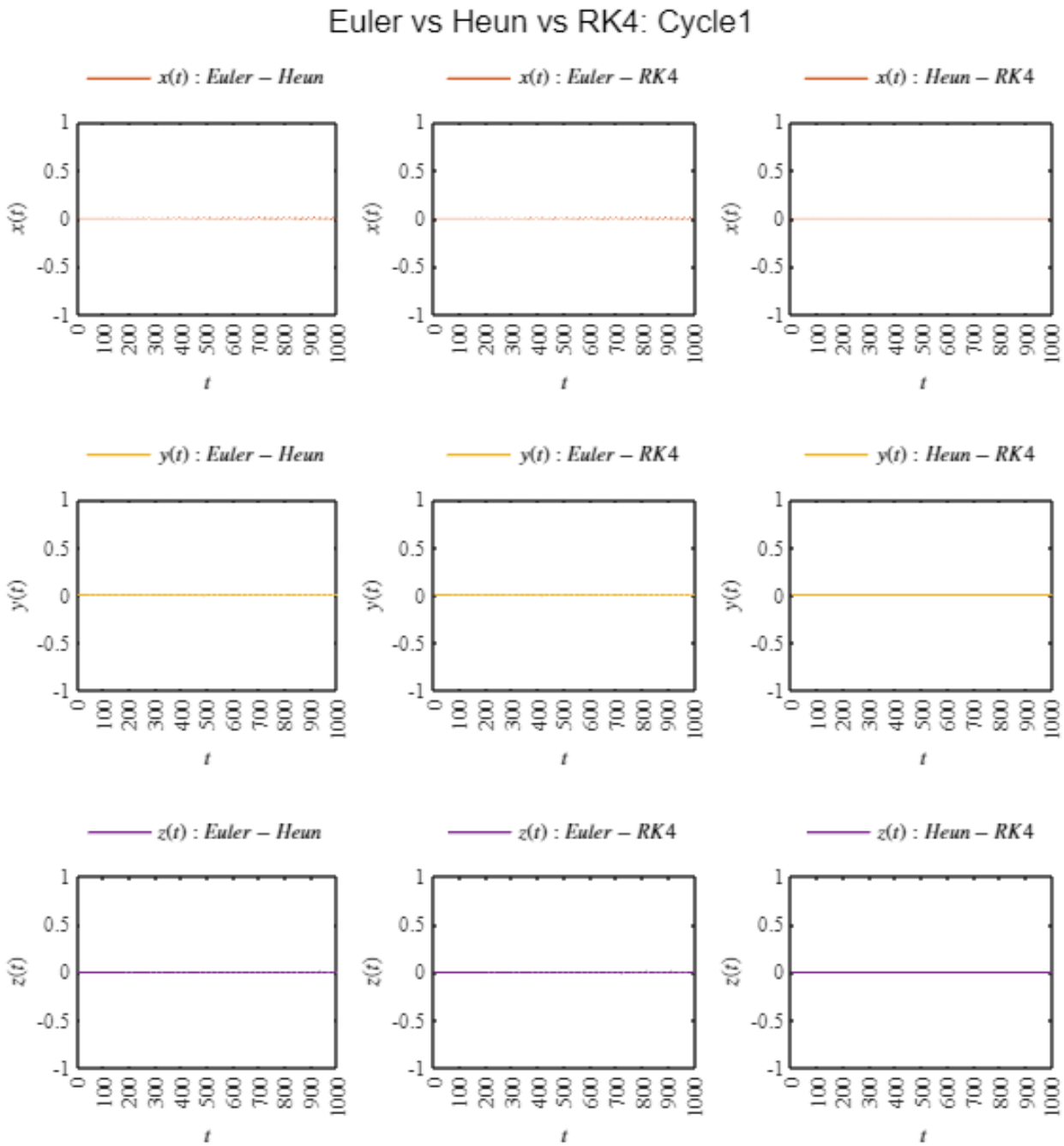
Comparación entre soluciones

```
plotdivergence(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Cycle1');...
    exportgraphics(gcf,'Solutions_Cycle1.pdf','ContentType','vector');
```



Error

```
ploterror(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Cycle1');...
    exportgraphics(gcf,'Error_Cycle1.pdf','ContentType','vector');
```



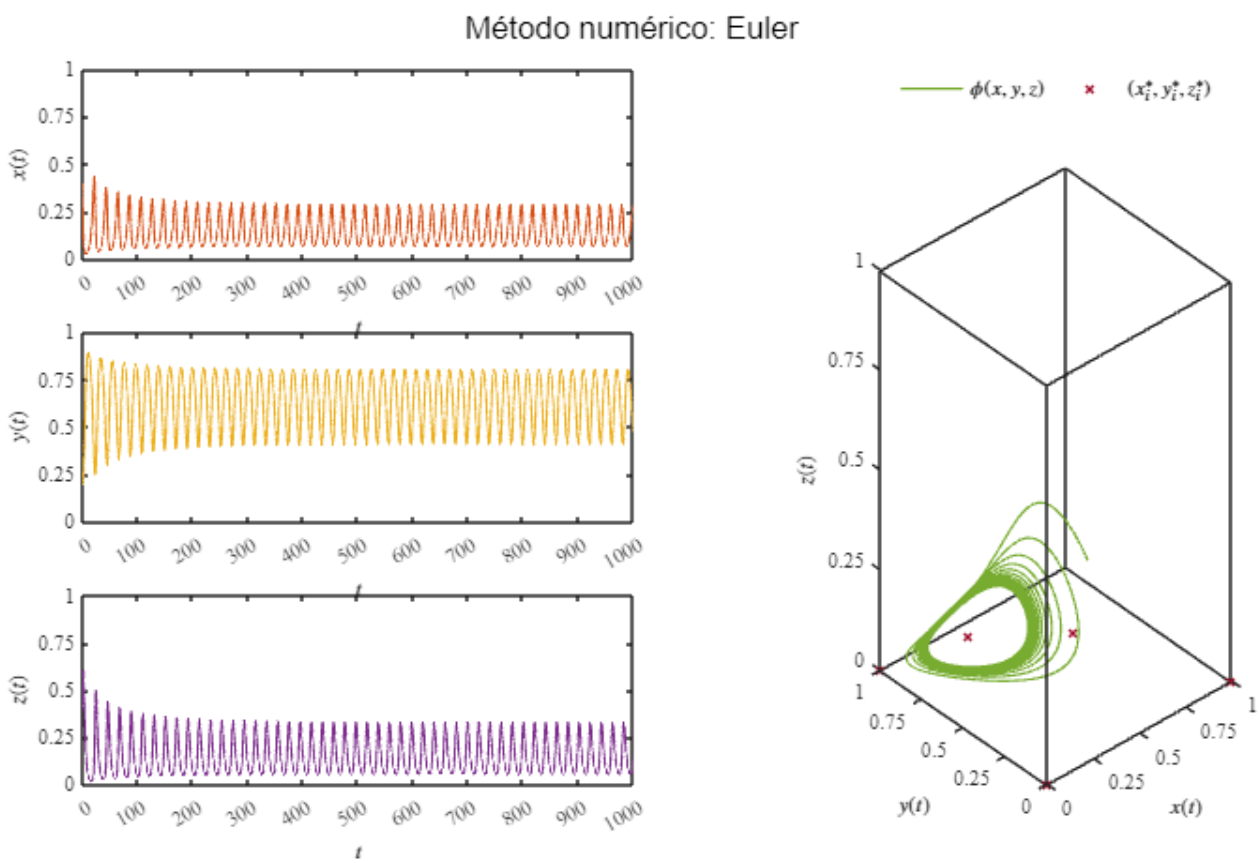
Comparación de métodos numéricos: Ciclo límite 2

```
D = getdata('cycle2'); f0 = D.f0; par = D.par;
dt = 1e-4; tend = 1000;
rhoi = zeros(round(tend/dt)+1,1);
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.481e-01 , 0.000e+00 , 3.407e-01)
eq2        ( 1.481e-01 , 6.296e-01 , 1.519e-01)
eq3        (-2.857e-01 , 1.714e+00 , 0.000e+00)
eq4        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq5        ( 1.000e+00 , 0.000e+00 , 0.000e+00)
eq6        ( 0.000e+00 , 0.000e+00 , 0.000e+00)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i +3.8e-01+0.0e+00i unstable
eq2        -5.3e-01+0.0e+00i +4.2e-03+3.3e-01i +4.2e-03-3.3e-01i unstable
eq3        -3.7e-01+3.5e-01i -3.7e-01-3.5e-01i -1.2e+00+0.0e+00i stable
eq4        -6.0e-01+0.0e+00i +2.5e-01+0.0e+00i -4.0e-01+0.0e+00i unstable
eq5        -1.0e+00+0.0e+00i -9.0e-01+0.0e+00i +2.3e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +6.0e-01+0.0e+00i +1.0e+00+0.0e+00i unstable
```

Euler

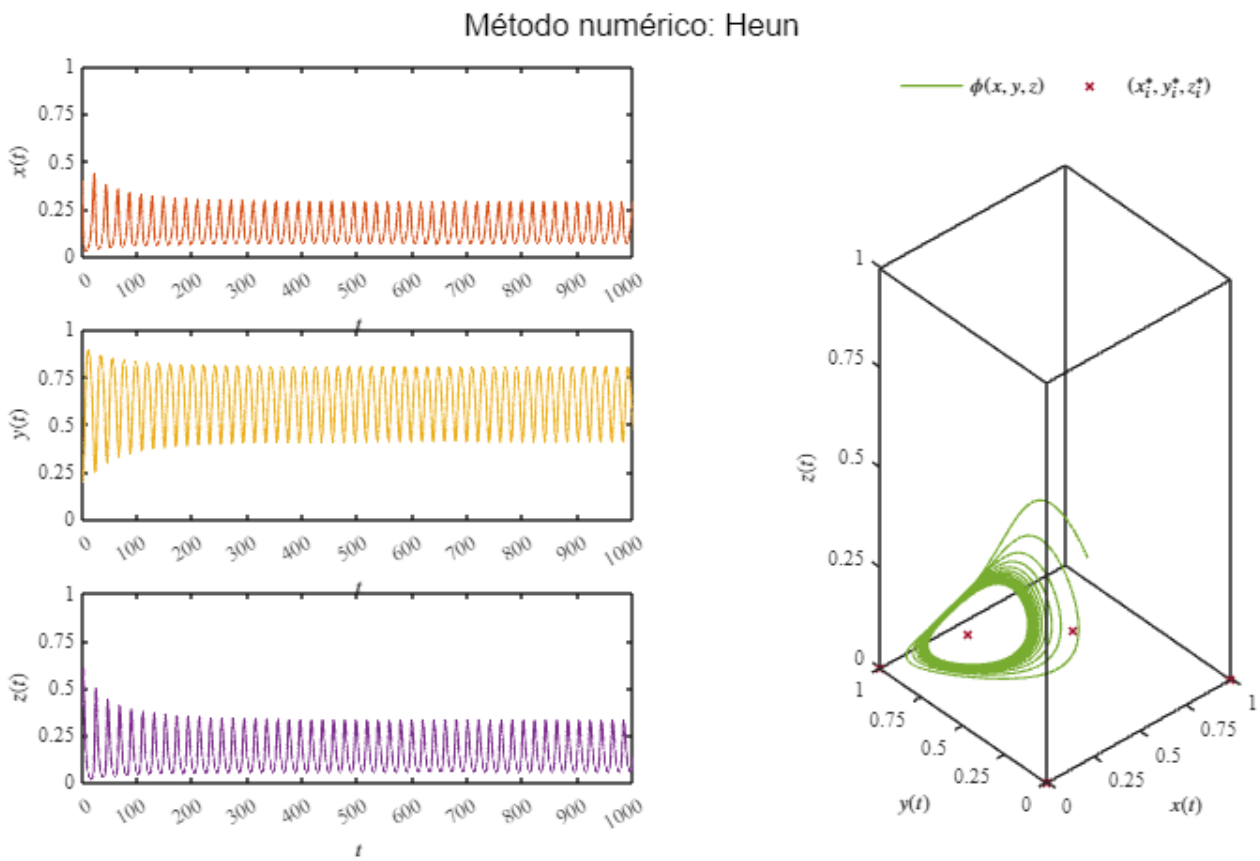
```
method = 'Euler';
[t,xe,ye,ze] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xe,ye,ze,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle2.pdf'],'ContentType','vector');
```



Heun

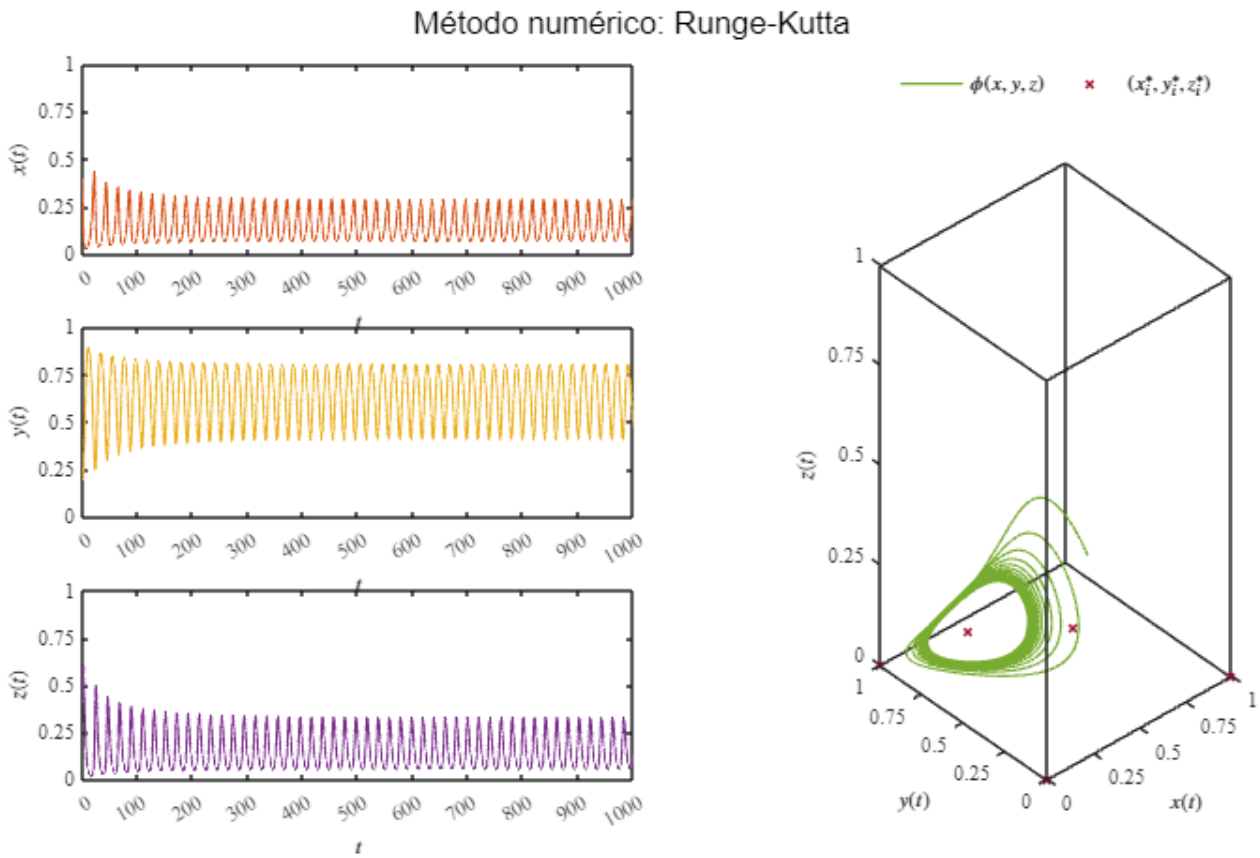
```
method = 'Heun';
[t,xh,yh,zh] = odesolver(method,f0,par,rhoi,tend,dt);
```

```
plotEDOS(t,xh,yh,zh,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle2.pdf'],'ContentType','vector');
```



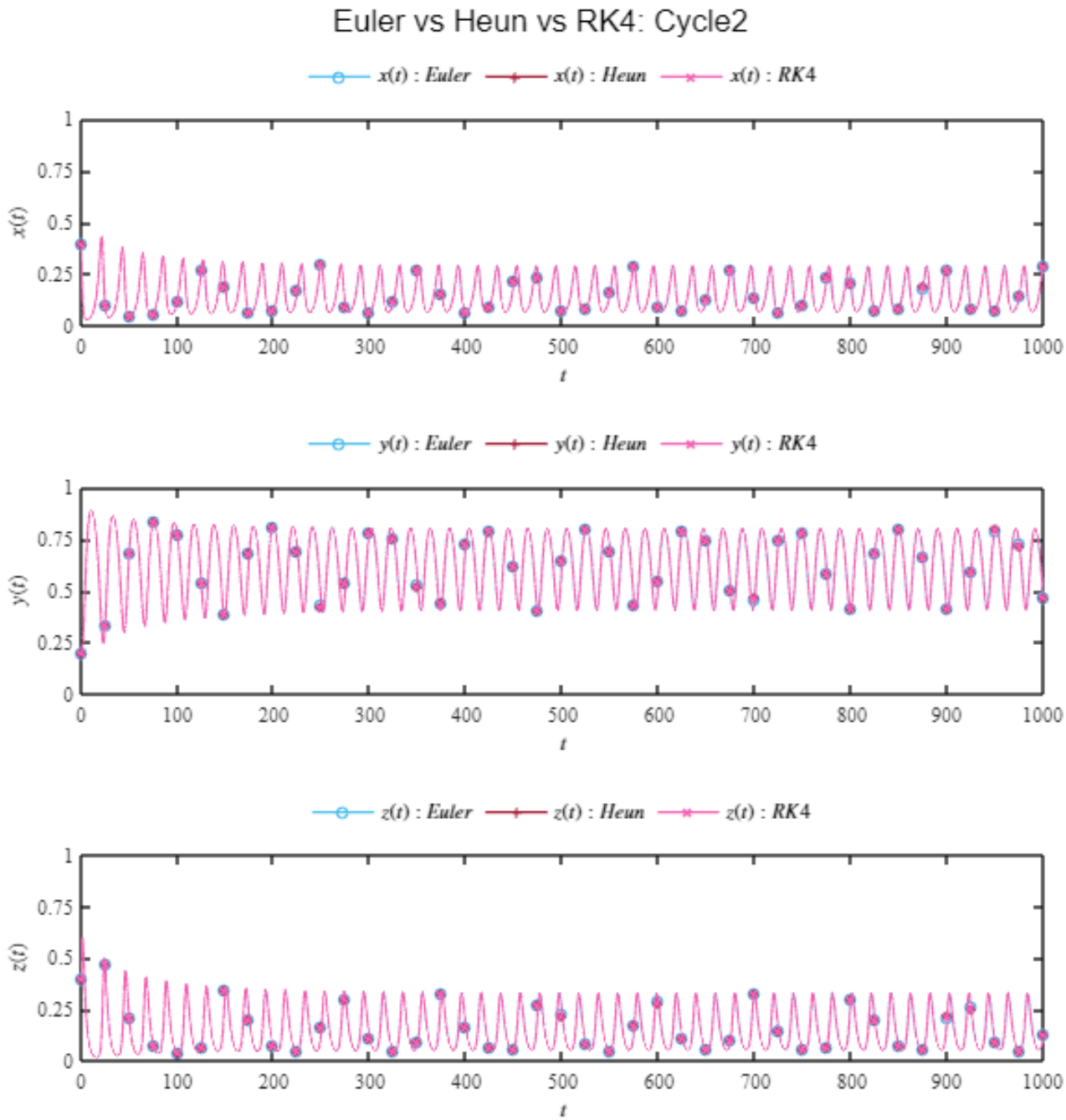
Runge-Kutta

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Cycle2.pdf'],'ContentType','vector');
```



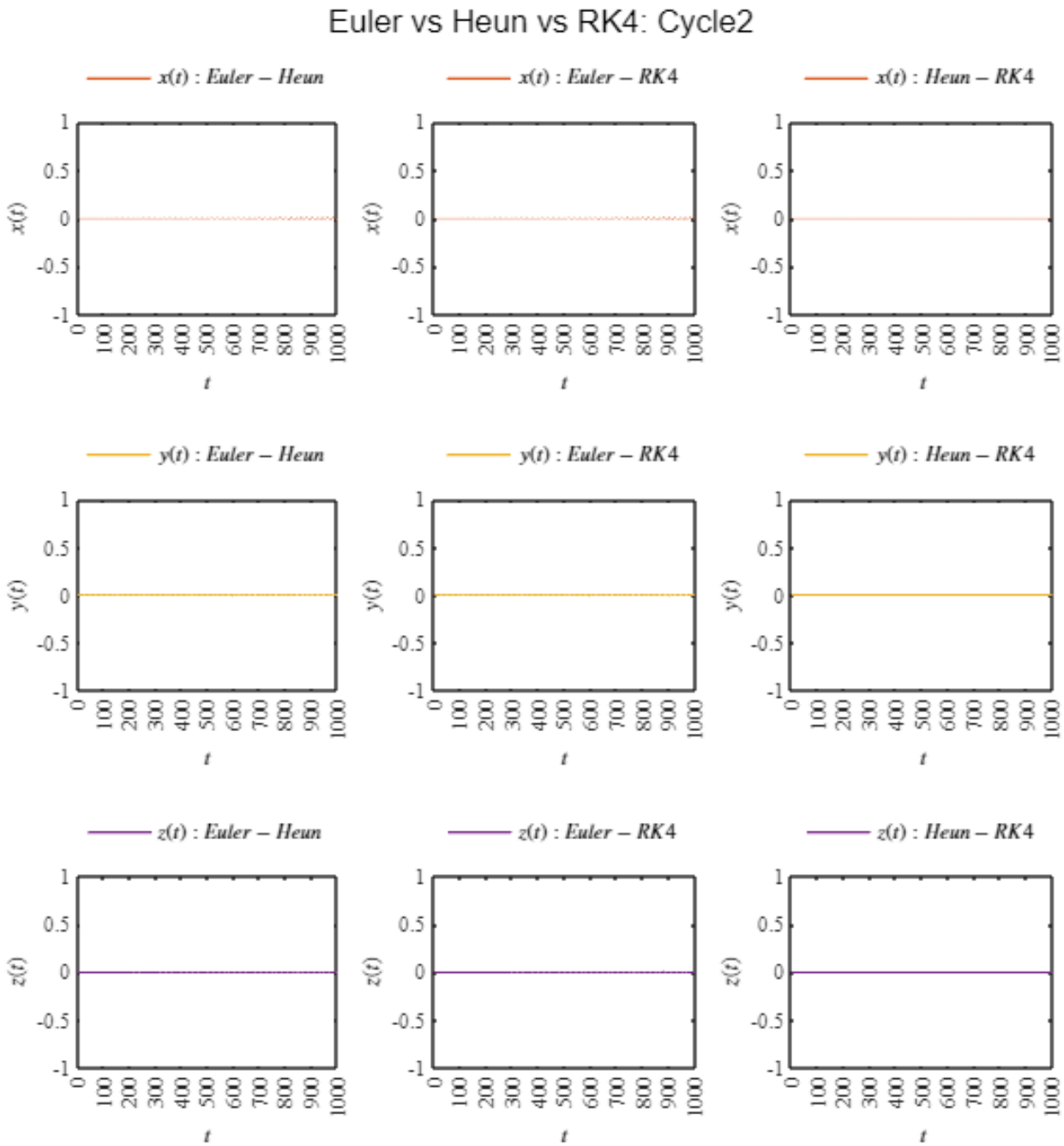
Comparación entre soluciones

```
plotdivergence(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Cycle2');...
    exportgraphics(gcf,'Solutions_Cycle2.pdf','ContentType','vector');
```



Error

```
ploterror(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Cycle2');...
    exportgraphics(gcf,'Error_Cycle2.pdf','ContentType','vector');
```



Comparación de métodos numéricos: Punto de Equilibrio

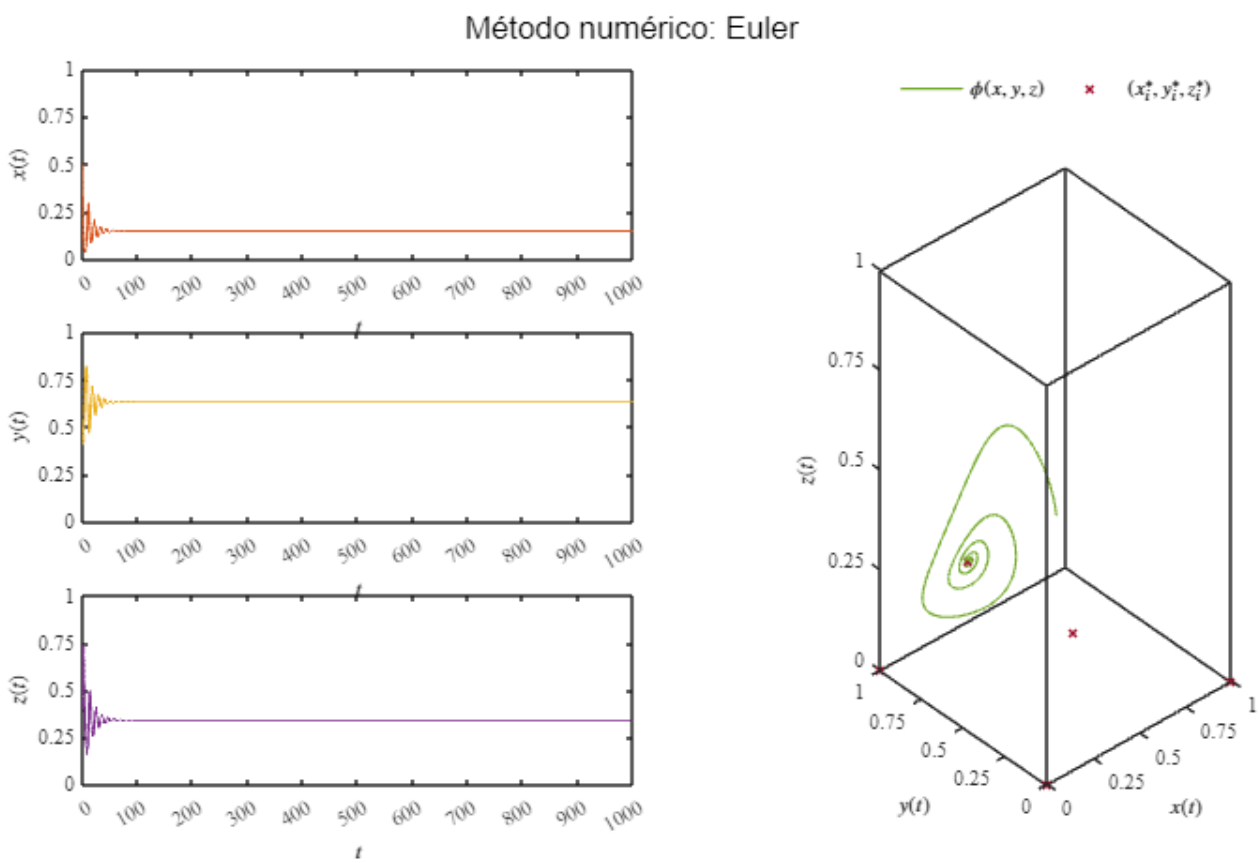
```
D = getdata('equilibrium'); f0 = D.f0; par = D.par;
dt = 1e-4; tend = 1000;

rhoi = zeros(round(tend/dt)+1,1);
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.481e-01 , 0.000e+00 , 3.407e-01)
eq2        ( 1.481e-01 , 6.296e-01 , 3.407e-01)
eq3        ( 1.000e+00 ,-1.500e+00 , 0.000e+00)
eq4        ( 0.000e+00 , 1.000e+00 , 0.000e+00)
eq5        ( 1.000e+00 , 0.000e+00 , 0.000e+00)
eq6        ( 0.000e+00 , 0.000e+00 , 0.000e+00)
```

```
Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i +3.8e-01+0.0e+00i unstable
eq2        -3.8e-01+0.0e+00i -7.4e-02+5.8e-01i -7.4e-02-5.8e-01i stable
eq3        +9.0e-01+0.0e+00i -1.0e+00+0.0e+00i +2.3e+00+0.0e+00i unstable
eq4        -6.0e-01+0.0e+00i +1.0e+00+0.0e+00i -4.0e-01+0.0e+00i unstable
eq5        -1.0e+00+0.0e+00i -9.0e-01+0.0e+00i +2.3e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +6.0e-01+0.0e+00i +1.0e+00+0.0e+00i unstable
```

Euler

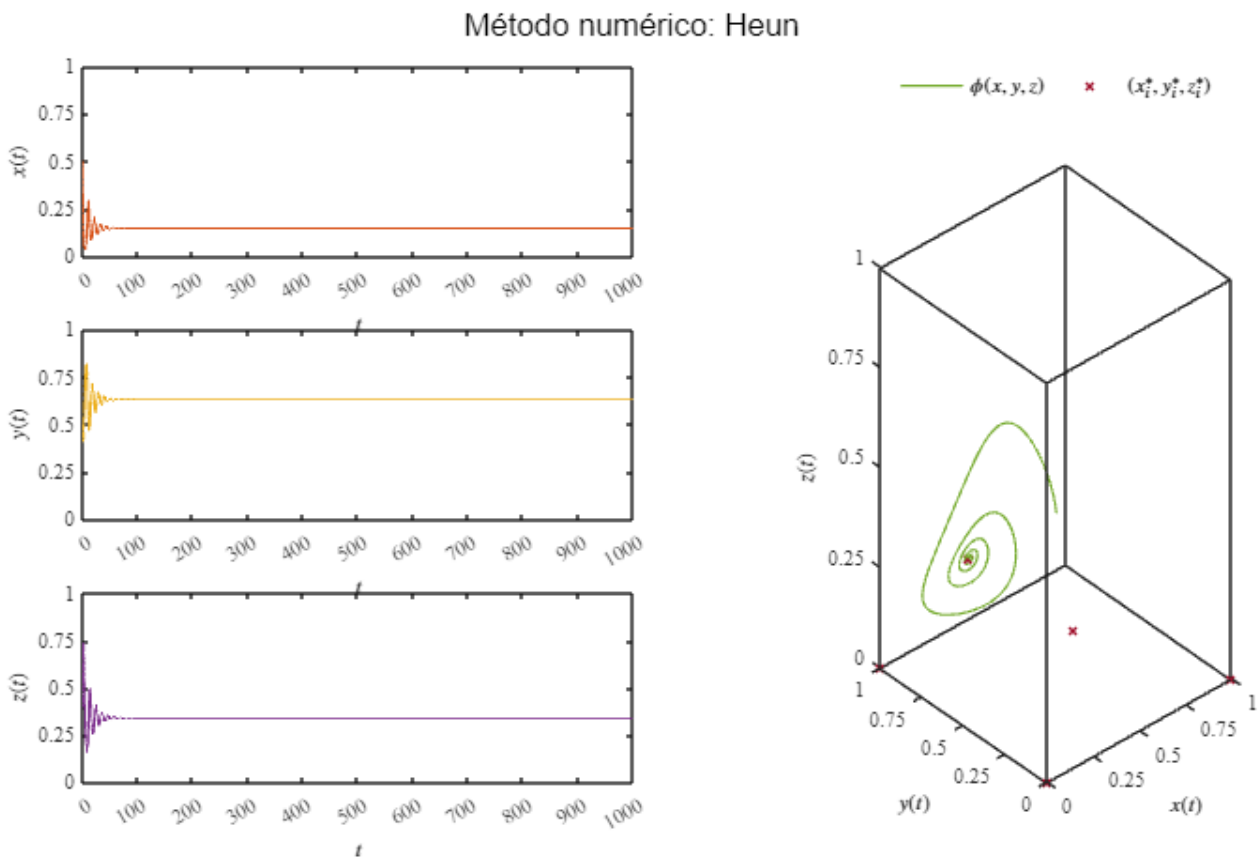
```
method = 'Euler';
[t,xe,ye,ze] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xe,ye,ze,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Equilibrium.pdf'],'ContentType','vector');
```



Heun

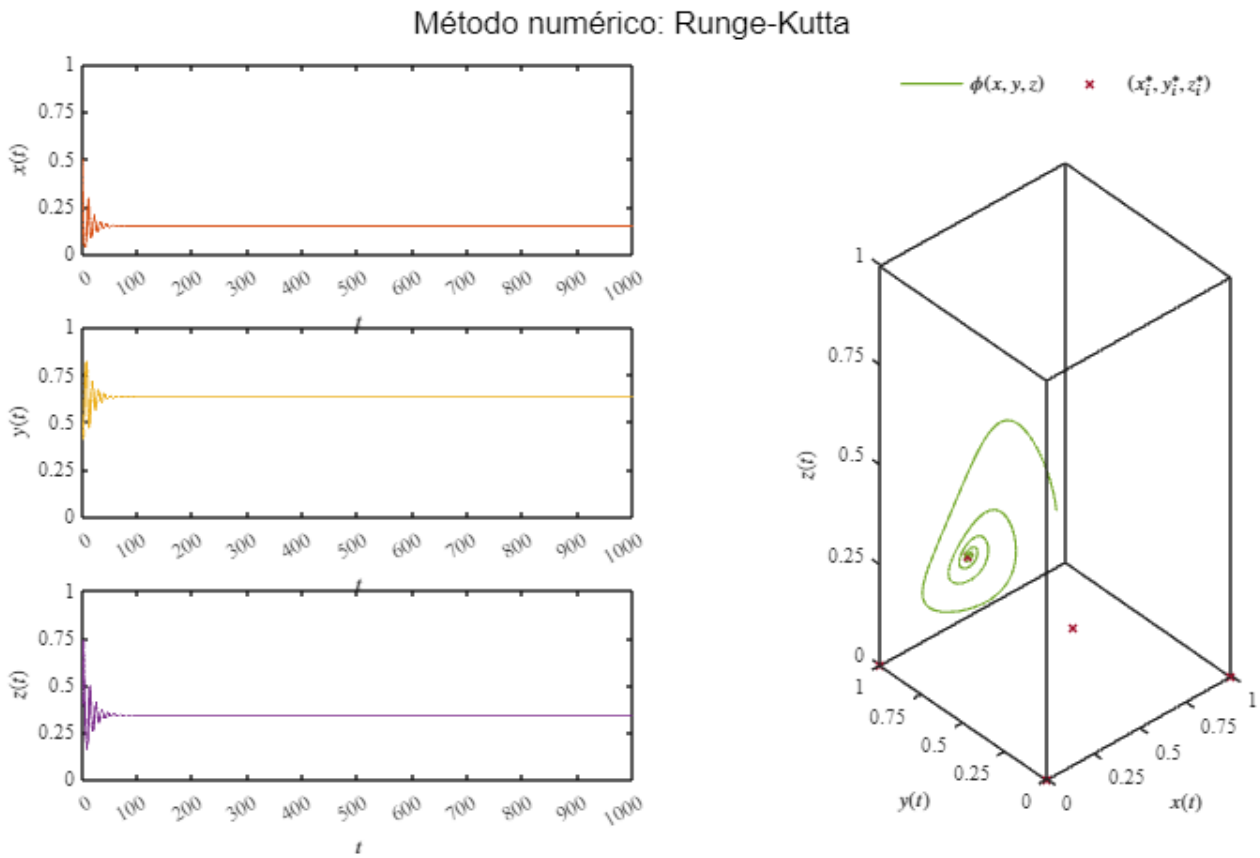
```
method = 'Heun';
[t,xh,yh,zh] = odesolver(method,f0,par,rhoi,tend,dt);
```

```
plotEDOS(t,xh,yh,zh,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Equilibrium.pdf'],'ContentType','vector');
```



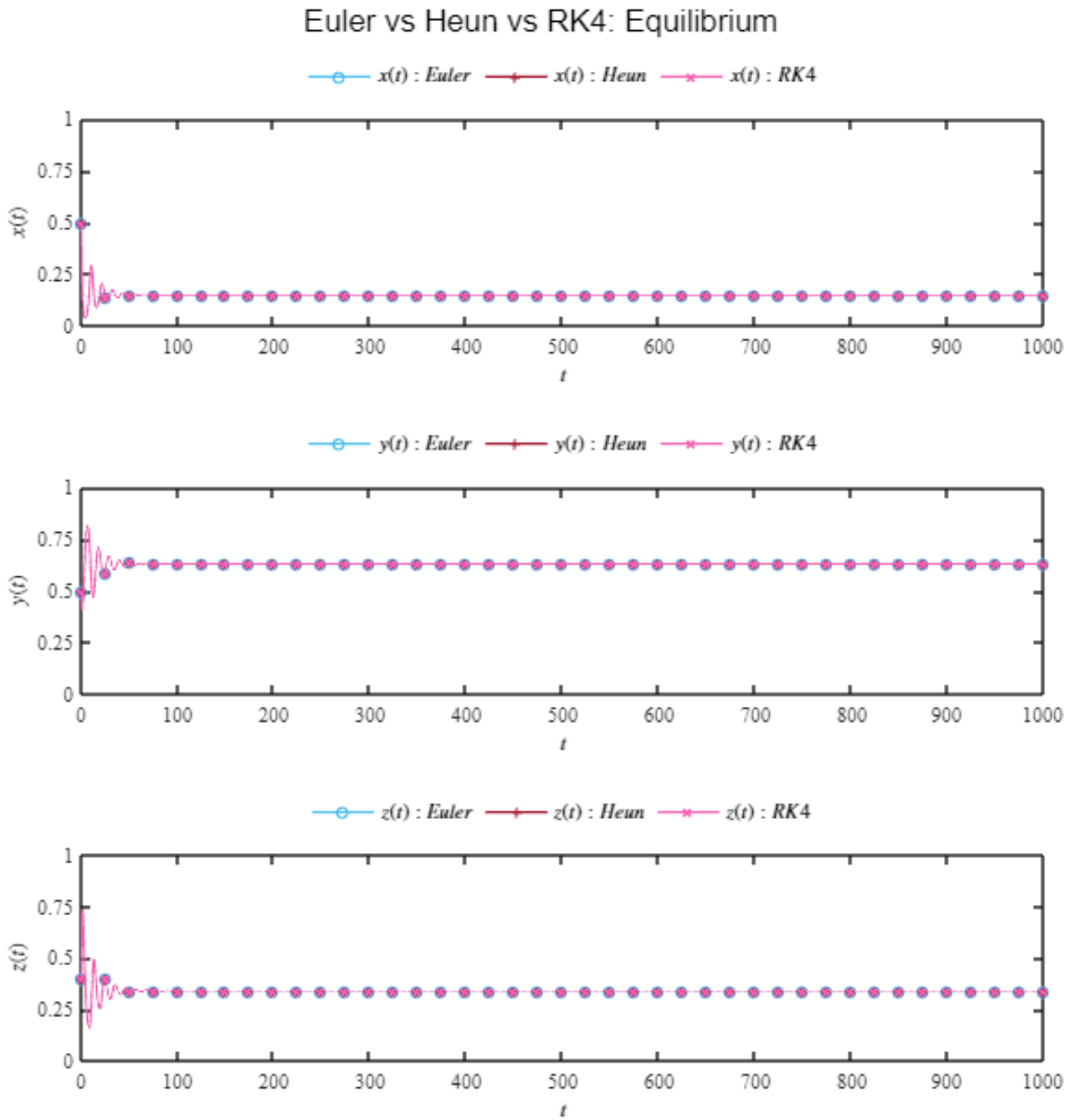
Runge-Kutta

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_Equilibrium.pdf'],'ContentType','vector');
```



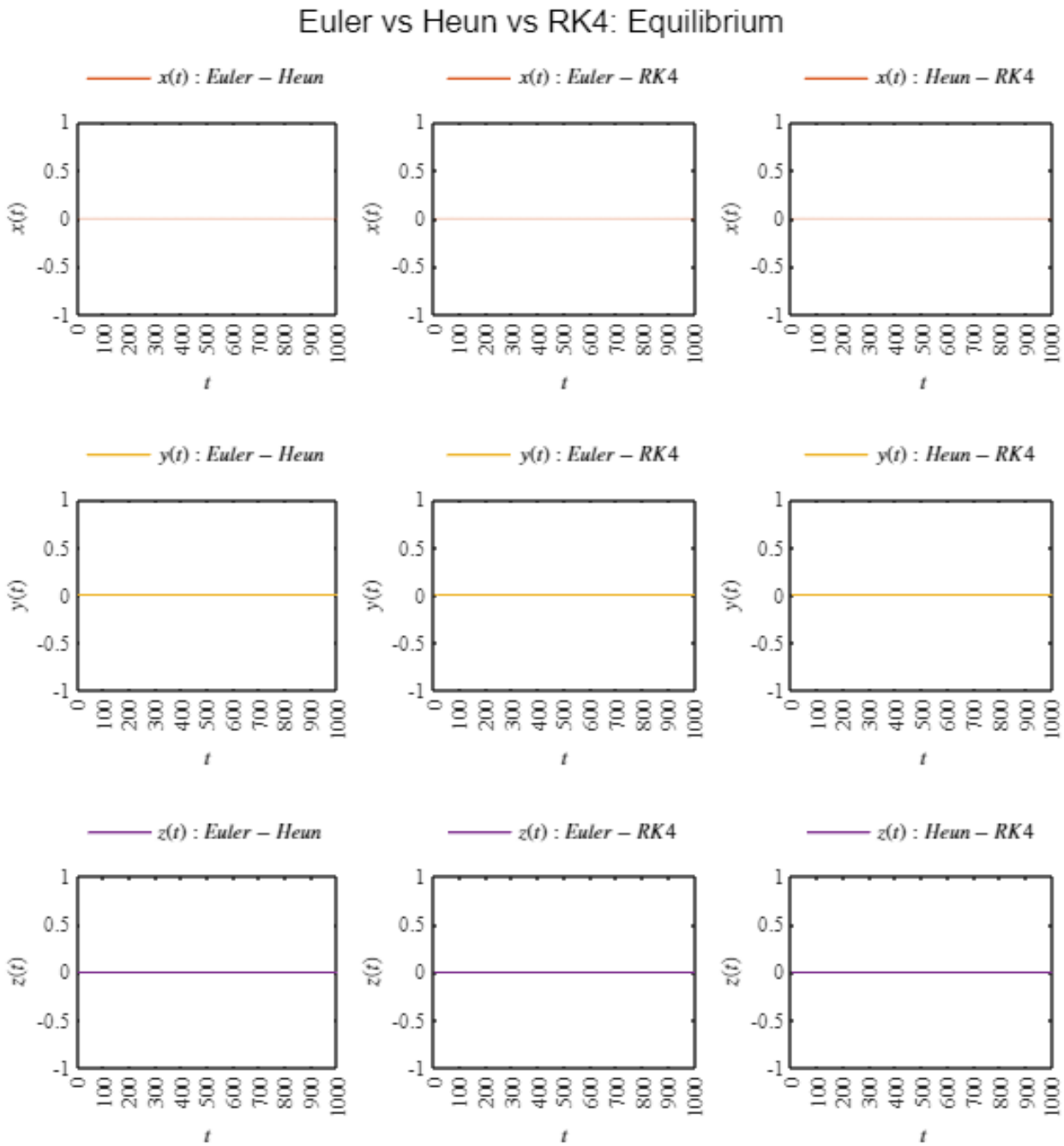
Comparación entre soluciones

```
plotdivergence(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Equilibrium');...
    exportgraphics(gcf,'Solutions_Equilibrium.pdf','ContentType','vector');
```



Error

```
ploterror(t,[xe,xh,xr],[ye,yh,yr],[ze,zh,zr]); sgtitle('Euler vs Heun vs RK4:
Equilibrium');...
    exportgraphics(gcf,'Error_Equilibrium.pdf','ContentType','vector');
```



Condiciones de estabilida asintótica

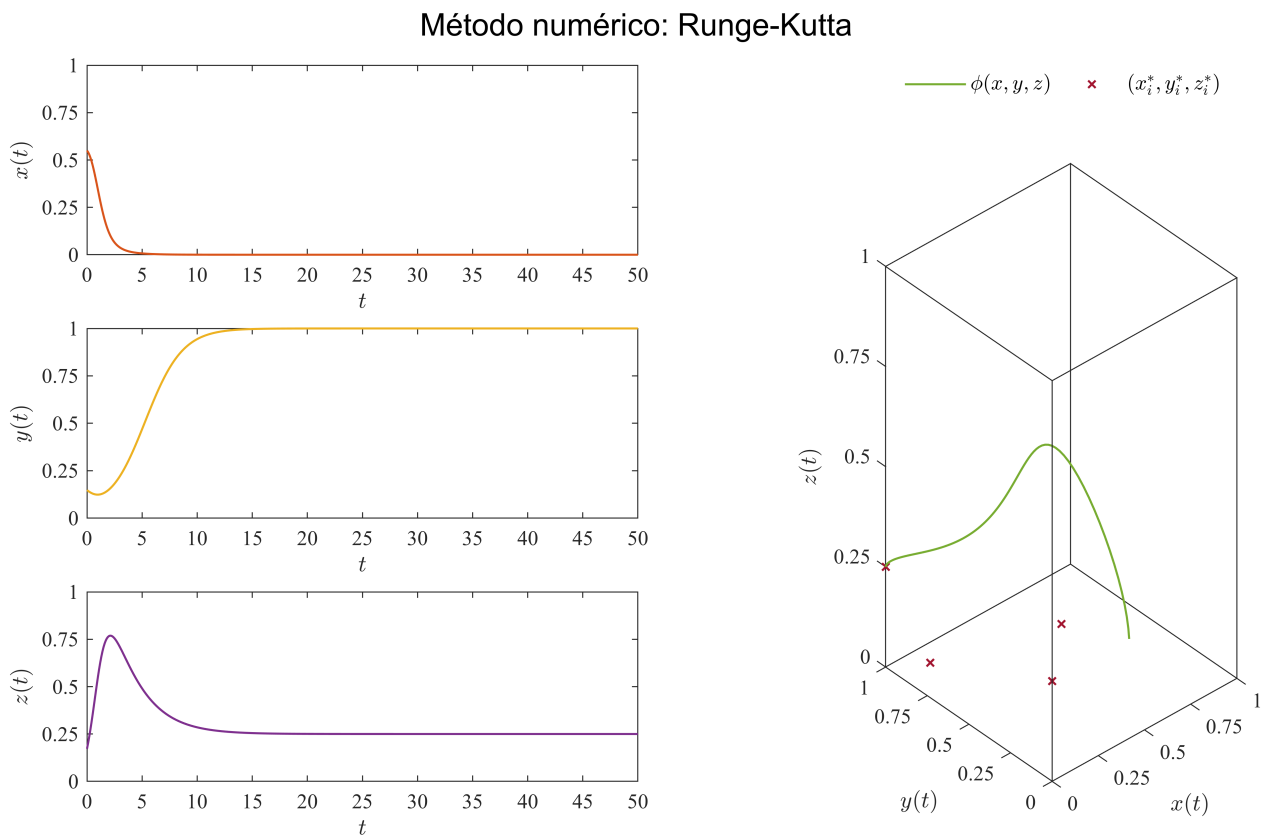
Aplicación del tratamiento: Atractor Caótico

```
D = getdata('chaos'); f0 = D.f0; par = D.par;
r1 = par(1); a12 = par(3); a13 = par(4); b2 = par(6); d3 = par(10);
CE = (r1 - a12/b2)*d3/a13*1.1; if CE == 0; CE = 0.1; end
dt = 1e-3; tend = 50;
```

```
rhoi = ones(round(tend/dt)+1,1)*CE;
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 5.062e-02 , 0.000e+00 , 3.798e-01)
eq2        ( 7.407e-02 , 8.148e-01 , 4.444e-02)
eq3        ( 7.407e-02 , 8.148e-01 , 4.444e-02)
eq4        ( 0.000e+00 , 1.000e+00 , 2.500e-01)
eq5        ( 1.098e+00 , 0.000e+00 ,-3.901e-02)
eq6        ( 0.000e+00 , 0.000e+00 , 2.500e-01)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -1.6e-01+3.4e-01i -1.6e-01-3.4e-01i +5.2e-01+0.0e+00i unstable
eq2        +4.5e-01-9.3e-02i -8.9e-01-4.6e-02i -3.3e-01-6.2e-01i unstable
eq3        +4.5e-01+9.3e-02i -8.9e-01+4.6e-02i -3.3e-01+6.2e-01i unstable
eq4        -4.0e-01+0.0e+00i -6.0e-01+0.0e+00i -6.2e-01+0.0e+00i stable
eq5        -1.2e+00+0.0e+00i +2.6e+00+0.0e+00i -1.0e+00+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +3.8e-01+0.0e+00i +6.0e-01+0.0e+00i unstable
```

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_rhoi_Chaos.pdf'],'ContentType','vector');
```



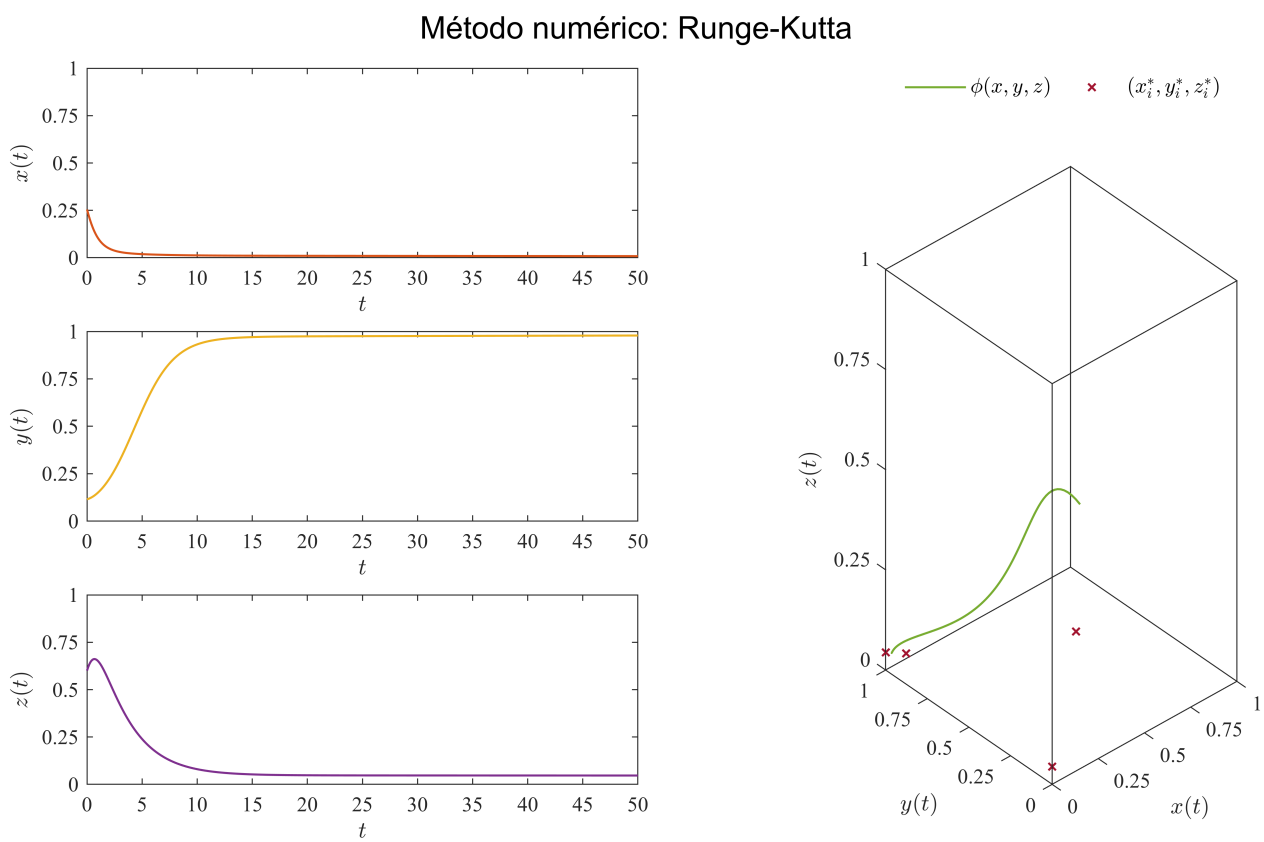
Aplicación del tratamiento: Órbita periódica

```
D = getdata('orbit'); f0 = D.f0; par = D.par;
```

```
r1 = par(1); a12 = par(3); a13 = par(4); b2 = par(6);d3 = par(10);
CE = (r1 - a12/b2)*d3/a13*1.1; if CE == 0; CE = 0.1; end
dt = 1e-3; tend = 50;
rhoi = ones(round(tend/dt)+1,1)*CE;
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.294e-01 , 0.000e+00 , 3.482e-01)
eq2        ( 3.407e-02 , 9.148e-01 , 5.704e-02)
eq3        ( 3.407e-02 , 9.148e-01 , 5.704e-02)
eq4        ( 0.000e+00 , 1.000e+00 , 4.400e-02)
eq5        ( 1.019e+00 , 0.000e+00 ,-7.488e-03)
eq6        ( 0.000e+00 , 0.000e+00 , 4.400e-02)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------------
eq1        -9.0e-02+5.5e-01i -9.0e-02-5.5e-01i +4.1e-01+0.0e+00i unstable
eq2        +7.4e-04+3.6e-03i -6.2e-01+1.2e-03i -2.8e-01-2.1e-02i unstable
eq3        +7.4e-04-3.6e-03i -6.2e-01-1.2e-03i -2.8e-01+2.1e-02i unstable
eq4        -4.0e-01+0.0e+00i -6.0e-01+0.0e+00i -1.0e-02+0.0e+00i stable
eq5        -1.0e+00+0.0e+00i +2.4e+00+0.0e+00i -9.3e-01+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +8.9e-01+0.0e+00i +6.0e-01+0.0e+00i unstable
```

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_rhoi_Orbit.pdf'],'ContentType','vector');
```

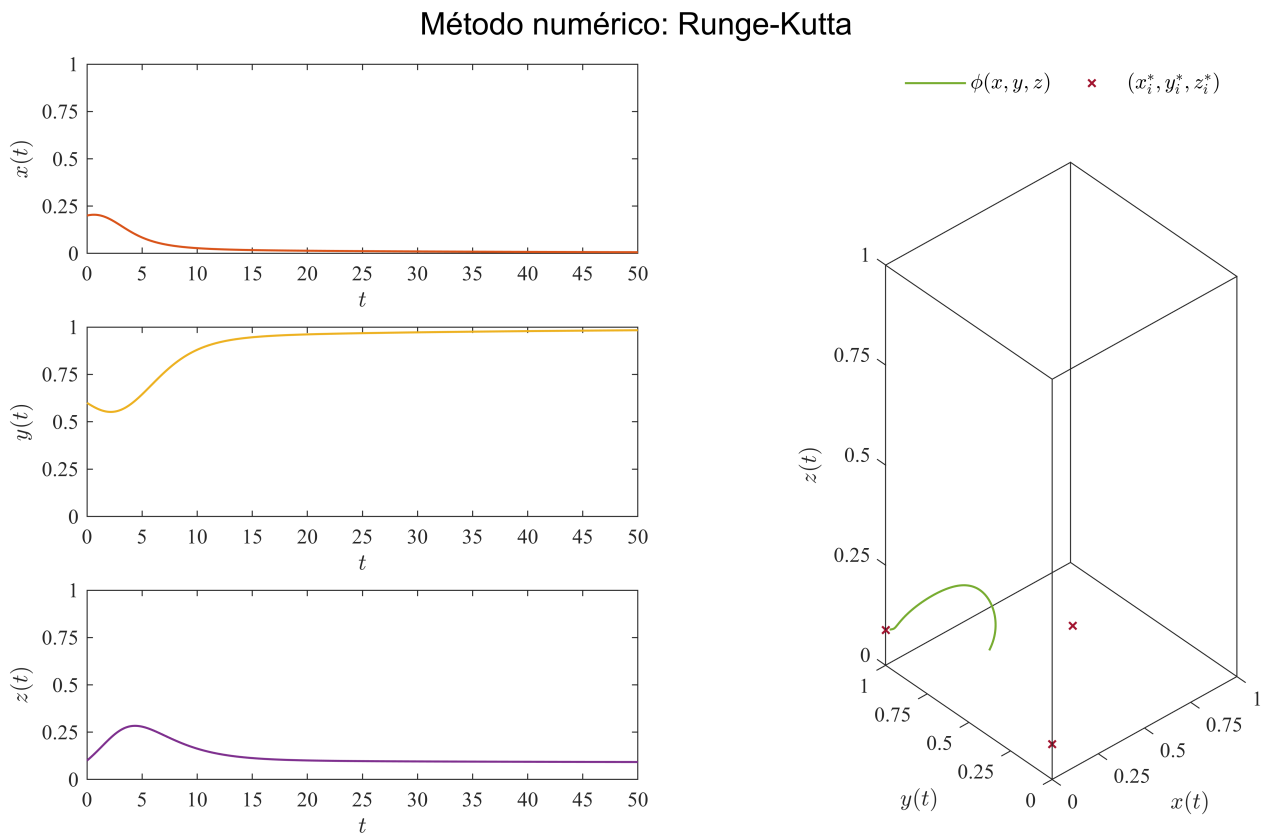


Aplicación del tratamiento: Ciclo límite 1

```
D = getdata('cycle1'); f0 = D.f0; par = D.par;
r1 = par(1); a12 = par(3); a13 = par(4); b2 = par(6); d3 = par(10);
CE = (r1 - a12/b2)*d3/a13*1.1; if CE == 0; CE = 0.1; end
dt = 1e-3; tend = 50;
rhoi = ones(round(tend/dt)+1,1)*CE;
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.115e-01 , 0.000e+00 , 3.554e-01)
eq2        (-2.593e-02 , 1.065e+00 , 6.963e-02)
eq3        (-2.593e-02 , 1.065e+00 , 6.963e-02)
eq4        ( 0.000e+00 , 1.000e+00 , 8.800e-02)
eq5        ( 1.037e+00 , 0.000e+00 ,-1.467e-02)
eq6        ( 0.000e+00 , 0.000e+00 , 8.800e-02)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------
eq1        -1.1e-01+5.2e-01i -1.1e-01-5.2e-01i +4.3e-01+0.0e+00i unstable
eq2        +2.2e-02-1.7e-02i -6.2e-01+3.2e-02i -4.9e-01-1.7e-01i unstable
eq3        +2.2e-02+1.7e-02i -6.2e-01-3.2e-02i -4.9e-01+1.7e-01i unstable
eq4        -4.0e-01+0.0e+00i -6.0e-01+0.0e+00i -2.0e-02+0.0e+00i stable
eq5        -1.1e+00+0.0e+00i +2.4e+00+0.0e+00i -9.6e-01+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +7.8e-01+0.0e+00i +6.0e-01+0.0e+00i unstable
```

```
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_rhoi_Cycle1.pdf'],'ContentType','vector');
```



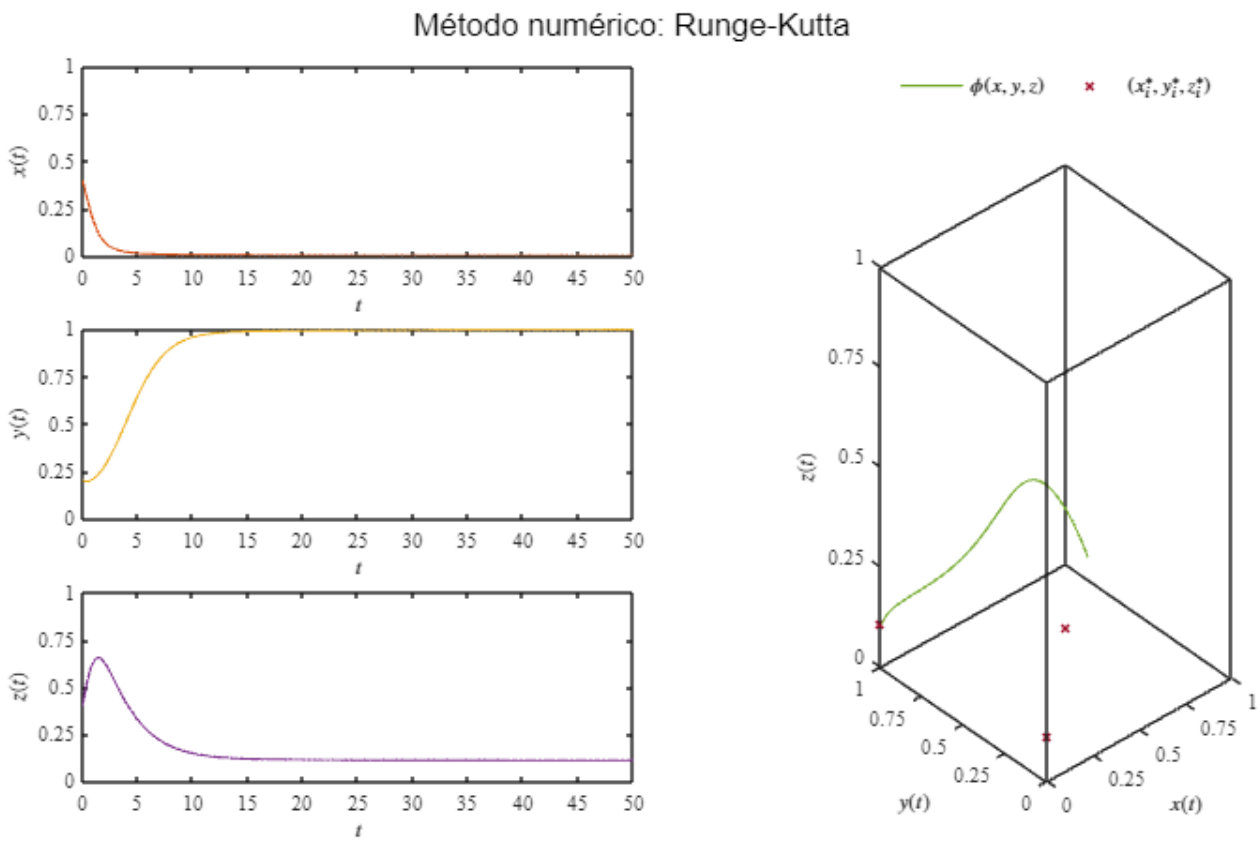
Aplicación del tratamiento: Ciclo límite 2

```
D = getdata('cycle2'); f0 = D.f0; par = D.par;
r1 = par(1); a12 = par(3); a13 = par(4); b2 = par(6); d3 = par(10);
CE = (r1 - a12/b2)*d3/a13*1.1; if CE == 0; CE = 0.1; end
dt = 1e-3; tend = 50;
rhoi = ones(round(tend/dt)+1,1)*CE;
E = equilibria(par,max(rhoi));
```

```
Equilibria  |  x*       |  y*       |  z*      |
-----------------------------------------------
eq1        ( 1.027e-01 , 0.000e+00 , 3.589e-01)
eq2        (-4.646e-02 , 1.116e+00 , 8.374e-02)
eq3        (-9.111e-02 , 1.228e+00 , 6.811e-02)
eq4        ( 0.000e+00 , 1.000e+00 , 1.100e-01)
eq5        ( 1.045e+00 , 0.000e+00 ,-1.816e-02)
eq6        ( 0.000e+00 , 0.000e+00 , 1.100e-01)


Equilibria |  lambda 1        |  lambda 2        |  lambda 3       | stability |
-------------------------------------------------------------------------------
eq1        -1.1e-01+5.0e-01i -1.1e-01-5.0e-01i +4.5e-01+0.0e+00i unstable
eq2        +9.7e-03+0.0e+00i -5.8e-01+6.5e-02i -5.8e-01-6.5e-02i unstable
eq3        -1.7e-02+0.0e+00i -6.4e-01+7.7e-02i -6.4e-01-7.7e-02i stable
eq4        -4.0e-01+0.0e+00i -6.0e-01+0.0e+00i -2.5e-02+0.0e+00i stable
eq5        -1.1e+00+0.0e+00i +2.5e+00+0.0e+00i -9.7e-01+0.0e+00i unstable
eq6        -4.0e-01+0.0e+00i +7.2e-01+0.0e+00i +6.0e-01+0.0e+00i unstable
```

```
method = 'Runge-Kutta';
```

```
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_rhoi_Cycle2.pdf'],'ContentType','vector');
```



Aplicación del tratamiento: Punto de equilibrio

```
D = getdata('equilibrium'); f0 = D.f0; par = D.par;
r1 = par(1); a12 = par(3); a13 = par(4); b2 = par(6); d3 = par(10);
CE = (r1 - a12/b2)*d3/a13*1.1; if CE == 0; CE = 0.1; end
dt = 1e-3; tend = 50;
rhoi = ones(round(tend/dt)+1,1)*CE;
E = equilibria(par,max(rhoi));
method = 'Runge-Kutta';
[t,xr,yr,zr] = odesolver(method,f0,par,rhoi,tend,dt);
plotEDOS(t,xr,yr,zr,E); sgtitle(['Método numérico: ',method]);...
    exportgraphics(gcf,[method,'_rhoi_Equilibrium.pdf'],'ContentType','vector');
```

Funciones

Parámetros y condiciones iniciales

```
function D = getdata(dynamics)
    r1 = 1; b1 = 1; a13 = 2.5;
    r2 = 0.6; b2 = 1; a21 = 1.5;
    r3 = 3; a31 = 0.3; d3 = 0.4;
```

```
    if dynamics == "chaos"
        a12 = 1;
        x0 = 0.55; y0 = 0.148; z0 = 0.172;
    elseif dynamics == "orbit"
        a12 = 0.9;
        x0 = 0.254781; y0 = 0.115167; z0 = 0.600527;
    elseif dynamics == "cycle1"
        a12 = 0.8;
        x0 = 0.2; y0 = 0.6; z0 = 0.1;
    elseif dynamics == "cycle2"
        a12 = 0.75;
        x0 = 0.4; y0 = 0.2; z0 = 0.4;
    elseif dynamics == "equilibrium"
        a12 = 0.;
        x0 = 0.5; y0 = 0.5; z0 = 0.4;
    end
    D.par = [r1 b1 a12 a13 r2 b2 a21 r3 a31 d3];
    D.f0 = [x0 y0 z0];
end
```

Métodos numéricos

```
function [t,x,y,z] = odesolver(method,f0,par,rhoi,tend,dt)
    method = lower(strtrim(method));
    r1 = par(1); b1 = par(2); a12 = par(3); a13 = par(4);
    r2 = par(5); b2 = par(6); a21 = par(7);
    r3 = par(8); a31 = par(9); d3 = par(10);

    n = round(tend/dt);
    t = (0:dt:tend)';
    x = zeros(length(t),1); x(1) = f0(1);
    y = zeros(length(t),1); y(1) = f0(2);
    z = zeros(length(t),1); z(1) = f0(3);

    for i = 1:n
        switch method
            case "euler"
                [fx,fy,fz] = f(x(i),y(i),z(i),rhoi(i));
                x(i+1) = x(i) + dt * fx;
                y(i+1) = y(i) + dt * fy;
                z(i+1) = z(i) + dt * fz;

            case "heun"
                [fx,fy,fz] = f(x(i),y(i),z(i),rhoi(i));
                xn = x(i) + dt*fx;
                yn = y(i) + dt*fy;
                zn = z(i) + dt*fz;
                [fxn,fyn,fzn] = f(xn,yn,zn,rhoi(i));
                x(i+1) = x(i) + (dt/2) * (fx + fxn);
                y(i+1) = y(i) + (dt/2) * (fy + fyn);
```

```
                z(i+1) = z(i) + (dt/2) * (fz + fzn);

            case {"rk4","runge-kutta"}
                [k1x,k1y,k1z] = f(x(i),y(i),z(i),rhoi(i));
                [k2x,k2y,k2z] = f(x(i)+(dt/2)*k1x,y(i)+(dt/2)*k1y, z(i)+(dt/
2)*k1z,rhoi(i));
                [k3x,k3y,k3z] = f(x(i)+(dt/2)*k2x,y(i)+(dt/2)*k2y, z(i)+(dt/
2)*k2z,rhoi(i));
                [k4x,k4y,k4z] = f(x(i)+dt*k3x, y(i)+dt*k3y, z(i)+dt*k3z,rhoi(i));
                x(i+1) = x(i) + (dt/6)*(k1x + 2*k2x + 2*k3x + k4x);
                y(i+1) = y(i) + (dt/6)*(k1y + 2*k2y + 2*k3y + k4y);
                z(i+1) = z(i) + (dt/6)*(k1z + 2*k2z + 2*k3z + k4z);

        end
    end

    function [dx,dy,dz] = f(x,y,z,rho)
        dx = r1*x*(1-b1*x) - a12*x*y - a13*x*z;
        dy = r2*y*(1-b2*y) - a21*x*y;
        dz = (r3 - a31)*x*z - d3*z +rho;
    end

    h = 10;
    t = t(1:1/(dt*h):end);
    x = x(1:1/(dt*h):end);
    y = y(1:1/(dt*h):end);
    z = z(1:1/(dt*h):end);
end
```

Soluciones y espacio de fase

```
function plotEDOS(t,x,y,z,E)
    set(figure(),'color','w')
    set(gcf,'units','centimeters','position',[1 1 25 15])
    FS = 10;
    c1 = [0.8500, 0.3250, 0.0980];
    c2 = [0.9290, 0.6940, 0.1250];
    c3 = [0.4940, 0.1840, 0.5560];
    c4 = [0.4660, 0.6740, 0.1880];
    c5 = [0.6350, 0.0780, 0.1840];

    subplot(3,6,[1 2 3]); box on; hold on;
    plot (t,x,'-','LineWidth',1,'Color',c1)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$x(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,6,[7 8 9]); box on; hold on;
```

```
    plot (t,y,'-','LineWidth',1,'Color',c2)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,6,[13 14 15]); box on; hold on;
    plot (t,z,'-','LineWidth',1,'Color',c3)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$z(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,6,[5 6 11 12 17 18]); hold on; grid off; box on; view([-42 21])
    ax = gca; ax.BoxStyle = 'full';
    plot3(x,y,z,'-','LineWidth',1,'Color',c4)
    plot3(E(:,1),E(:,2),E(:,3),'x','MarkerSize',5,'LineWidth',1,'Color',c5)
    L = legend('$\phi(x,y,z)$','$(x^*_i,y^*_i,z^*_i)$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 1]); xticks(0:0.25:1)
    ylim([0 1]); yticks(0:0.25:1)
    zlim([0 1]); zticks(0:0.25:1)
    xlabel('$x(t)$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    zlabel('$z(t)$','Interpreter','latex','FontSize',FS)
    set(gca,'FontName','Times New Roman','FontSize',FS)
end

function plotdivergence(t,x,y,z)
    set(figure(),'color','w')
    set(gcf,'units','centimeters','position',[1 1 25 25])
    FS = 10;
    c5 = [0.3010, 0.7450, 0.9330];
    c6 = [0.6350, 0.0780, 0.1840];
    c7 = [0.9490, 0.3765, 0.7020];
    colororder([c5;c6;c7]);


    subplot(3,1,1); box on; hold on; grid off;
    plot(t,x(:,1),'-o',t,x(:,2),'-+',t,x(:,3),'-x',...
        'LineWidth',1,'MarkerSize',5,'MarkerIndices',1:250:length(t));
    L = legend('$x(t)$: Euler$','$x(t)$: Heun$','$x(t)$: RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$x(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
```

```
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,1,2); box on; hold on; grid off;
    plot(t,y(:,1),'-o',t,y(:,2),'-+',t,y(:,3),'-x',...
        'LineWidth',1,'MarkerSize',5,'MarkerIndices',1:250:length(t));
    L = legend('$y(t): Euler$','$y(t): Heun$','$y(t): RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,1,3); box on; hold on; grid off;
    plot(t,z(:,1),'-o',t,z(:,2),'-+',t,z(:,3),'-x',...
        'LineWidth',1,'MarkerSize',5,'MarkerIndices',1:250:length(t));
    L = legend('$z(t): Euler$','$z(t): Heun$','$z(t): RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$z(t)$','Interpreter','latex','FontSize',FS)
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([0 1]); yticks(0:0.25:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)
end

function ploterror(t,x,y,z)
    set(figure(),'color','w')
    set(gcf,'units','centimeters','position',[1 1 25 25])
    FS = 10;
    c1 = [0.8500, 0.3250, 0.0980];
    c2 = [0.9290, 0.6940, 0.1250];
    c3 = [0.4940, 0.1840, 0.5560];


    subplot(3,3,1); box on; hold on;
    plot (t,x(:,1) - x(:,2),'-','LineWidth',1,'Color',c1)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$x(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$x(t): Euler - Heun$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,2); box on; hold on;
    plot (t,x(:,1) - x(:,3),'-','LineWidth',1,'Color',c1)
```

```
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$x(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$x(t): Euler - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,3); box on; hold on;
    plot (t,x(:,2) - x(:,3),'-','LineWidth',1,'Color',c1)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$x(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$x(t): Heun - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,4); box on; hold on;
    plot (t,y(:,1) - y(:,2),'-','LineWidth',1,'Color',c2)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$y(t): Euler - Heun$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,5); box on; hold on;
    plot (t,y(:,1) - y(:,3),'-','LineWidth',1,'Color',c2)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$y(t): Euler - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,6); box on; hold on;
    plot (t,y(:,2) - y(:,3),'-','LineWidth',1,'Color',c2)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$y(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$y(t): Heun - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
```

```
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,7); box on; hold on;
    plot (t,z(:,1) - z(:,2),'-','LineWidth',1,'Color',c3)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$z(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$z(t): Euler - Heun$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,8); box on; hold on;
    plot (t,z(:,1) - z(:,3),'-','LineWidth',1,'Color',c3)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$z(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$z(t): Euler - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

    subplot(3,3,9); box on; hold on;
    plot (t,z(:,2) - z(:,3),'-','LineWidth',1,'Color',c3)
    xlabel('$t$','Interpreter','latex','FontSize',FS)
    ylabel('$z(t)$','Interpreter','latex','FontSize',FS)
    L = legend('$z(t): Heun - RK4$');
    set(L,'Interpreter','Latex','FontSize',FS,'Location','NorthOutside',...
        'Box','Off','Orientation','Horizontal');
    xlim([0 inf]); xticks(0:t(end)/10:t(end))
    ylim([-1 1]); yticks(-1:0.5:1)
    set(gca,'FontName','Times New Roman','FontSize',FS)

end
```

Equilibrios

```
function [E,lambda] = equilibria(par,rhoi)
r1 = par(1); b1 = par(2); a12 = par(3); a13 = par(4);
r2 = par(5); b2 = par(6); a21 = par(7);
r3 = par(8); a31 = par(9); d3 = par(10);

e1 = [(r1 - (a13*(a31*r1 - r1*r3 - sqrt(r1*(r1*a31^2 + 2*r1*a31*b1*d3 -
4*a13*rhoi*a31*b1 - 2*r1*a31*r3 + r1*b1^2*d3^2 - 2*r1*b1*d3*r3 + 4*a13*rhoi*b1*r3
+ r1*r3^2)) + b1*d3*r1))/(2*(a13*a31 - a13*r3)))/(b1*r1); 0; (a31*r1 - r1*r3 -
```

```
sqrt(r1*(r1*a31^2 + 2*r1*a31*b1*d3 - 4*a13*rhoi*a31*b1 - 2*r1*a31*r3 + r1*b1^2*d3^2
- 2*r1*b1*d3*r3 + 4*a13*rhoi*b1*r3 + r1*r3^2)) + b1*d3*r1)/(2*(a13*a31 - a13*r3))];
e2 = [(r2*(a12 - b2*r1))/(a12*a21 - b1*b2*r1*r2) - (sqrt(a12^2*a21^2*d3^2
+ 2*a12^2*a21*a31*d3*r2 - 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2
- 2*a12^2*a31*r2^2*r3 + a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2
+ 4*a13*rhoi*a12*a21*a31*b2*r2 - 2*a12*a21*b1*b2*d3^2*r1*r2 +
2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3 - 2*a12*a31^2*b2*r1*r2^2 -
2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3 + 2*a12*b1*b2*d3*r1*r2^2*r3
- 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2 + 2*a31*b1*b2^2*d3*r1^2*r2^2 -
4*a13*rhoi*a31*b1*b2^2*r1*r2^2 - 2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2
- 2*b1*b2^2*d3*r1^2*r2^2*r3 + 4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2)
+ a12*a21*d3 + a12*a31*r2 - a12*r2*r3 - a31*b2*r1*r2 + b2*r1*r2*r3 -
b1*b2*d3*r1*r2)/(2*(a12*a21 - b1*b2*r1*r2)*(a31 - r3)); (a21*r1 - b1*r1*r2)/
(a12*a21 - b1*b2*r1*r2) + (a21*(sqrt(a12^2*a21^2*d3^2 + 2*a12^2*a21*a31*d3*r2
- 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2 - 2*a12^2*a31*r2^2*r3 +
a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2 + 4*a13*rhoi*a12*a21*a31*b2*r2 -
2*a12*a21*b1*b2*d3^2*r1*r2 + 2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3
- 2*a12*a31^2*b2*r1*r2^2 - 2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3
+ 2*a12*b1*b2*d3*r1*r2^2*r3 - 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2
+ 2*a31*b1*b2^2*d3*r1^2*r2^2 - 4*a13*rhoi*a31*b1*b2^2*r1*r2^2 -
2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2 - 2*b1*b2^2*d3*r1^2*r2^2*r3 +
4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2) + a12*a21*d3 + a12*a31*r2
- a12*r2*r3 - a31*b2*r1*r2 + b2*r1*r2*r3 - b1*b2*d3*r1*r2))/(2*b2*r2*(a12*a21
- b1*b2*r1*r2)*(a31 - r3)); -(sqrt(a12^2*a21^2*d3^2 + 2*a12^2*a21*a31*d3*r2
- 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2 - 2*a12^2*a31*r2^2*r3 +
a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2 + 4*a13*rhoi*a12*a21*a31*b2*r2 -
2*a12*a21*b1*b2*d3^2*r1*r2 + 2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3
- 2*a12*a31^2*b2*r1*r2^2 - 2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3
+ 2*a12*b1*b2*d3*r1*r2^2*r3 - 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2
+ 2*a31*b1*b2^2*d3*r1^2*r2^2 - 4*a13*rhoi*a31*b1*b2^2*r1*r2^2 -
2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2 - 2*b1*b2^2*d3*r1^2*r2^2*r3 +
4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2) + a12*a21*d3 + a12*a31*r2 -
a12*r2*r3 - a31*b2*r1*r2 + b2*r1*r2*r3 - b1*b2*d3*r1*r2)/(2*a13*b2*r2*(a31 - r3))];
e3 = [(r2*(a12 - b2*r1))/(a12*a21 - b1*b2*r1*r2) + (sqrt(a12^2*a21^2*d3^2
+ 2*a12^2*a21*a31*d3*r2 - 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2
- 2*a12^2*a31*r2^2*r3 + a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2
+ 4*a13*rhoi*a12*a21*a31*b2*r2 - 2*a12*a21*b1*b2*d3^2*r1*r2 +
2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3 - 2*a12*a31^2*b2*r1*r2^2 -
2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3 + 2*a12*b1*b2*d3*r1*r2^2*r3
- 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2 + 2*a31*b1*b2^2*d3*r1^2*r2^2 -
4*a13*rhoi*a31*b1*b2^2*r1*r2^2 - 2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2
- 2*b1*b2^2*d3*r1^2*r2^2*r3 + 4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2)
- a12*a21*d3 - a12*a31*r2 + a12*r2*r3 + a31*b2*r1*r2 - b2*r1*r2*r3 +
b1*b2*d3*r1*r2)/(2*(a12*a21 - b1*b2*r1*r2)*(a31 - r3)); (a21*r1 - b1*r1*r2)/
(a12*a21 - b1*b2*r1*r2) - (a21*(sqrt(a12^2*a21^2*d3^2 + 2*a12^2*a21*a31*d3*r2
- 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2 - 2*a12^2*a31*r2^2*r3 +
a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2 + 4*a13*rhoi*a12*a21*a31*b2*r2 -
2*a12*a21*b1*b2*d3^2*r1*r2 + 2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3
- 2*a12*a31^2*b2*r1*r2^2 - 2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3
+ 2*a12*b1*b2*d3*r1*r2^2*r3 - 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2
```

```
+ 2*a31*b1*b2^2*d3*r1^2*r2^2 - 4*a13*rhoi*a31*b1*b2^2*r1*r2^2 -
2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2 - 2*b1*b2^2*d3*r1^2*r2^2*r3 +
4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2) - a12*a21*d3 - a12*a31*r2
+ a12*r2*r3 + a31*b2*r1*r2 - b2*r1*r2*r3 + b1*b2*d3*r1*r2))/(2*b2*r2*(a12*a21
- b1*b2*r1*r2)*(a31 - r3)); (sqrt(a12^2*a21^2*d3^2 + 2*a12^2*a21*a31*d3*r2
- 2*a12^2*a21*d3*r2*r3 + a12^2*a31^2*r2^2 - 2*a12^2*a31*r2^2*r3 +
a12^2*r2^2*r3^2 - 2*a12*a21*a31*b2*d3*r1*r2 + 4*a13*rhoi*a12*a21*a31*b2*r2 -
2*a12*a21*b1*b2*d3^2*r1*r2 + 2*a12*a21*b2*d3*r1*r2*r3 - 4*a13*rhoi*a12*a21*b2*r2*r3
- 2*a12*a31^2*b2*r1*r2^2 - 2*a12*a31*b1*b2*d3*r1*r2^2 + 4*a12*a31*b2*r1*r2^2*r3
+ 2*a12*b1*b2*d3*r1*r2^2*r3 - 2*a12*b2*r1*r2^2*r3^2 + a31^2*b2^2*r1^2*r2^2
+ 2*a31*b1*b2^2*d3*r1^2*r2^2 - 4*a13*rhoi*a31*b1*b2^2*r1*r2^2 -
2*a31*b2^2*r1^2*r2^2*r3 + b1^2*b2^2*d3^2*r1^2*r2^2 - 2*b1*b2^2*d3*r1^2*r2^2*r3 +
4*a13*rhoi*b1*b2^2*r1*r2^2*r3 + b2^2*r1^2*r2^2*r3^2) - a12*a21*d3 - a12*a31*r2 +
a12*r2*r3 + a31*b2*r1*r2 - b2*r1*r2*r3 + b1*b2*d3*r1*r2)/(2*a13*b2*r2*(a31 - r3))];
e4 = [0; 1/b2; rhoi/d3];
e5 = [(r1 - (a13*(a31*r1 - r1*r3 + sqrt(r1*(r1*a31^2 + 2*r1*a31*b1*d3 -
4*a13*rhoi*a31*b1 - 2*r1*a31*r3 + r1*b1^2*d3^2 - 2*r1*b1*d3*r3 + 4*a13*rhoi*b1*r3
+ r1*r3^2)) + b1*d3*r1))/(2*(a13*a31 - a13*r3)))/(b1*r1); 0; (a31*r1 - r1*r3 +
sqrt(r1*(r1*a31^2 + 2*r1*a31*b1*d3 - 4*a13*rhoi*a31*b1 - 2*r1*a31*r3 + r1*b1^2*d3^2
- 2*r1*b1*d3*r3 + 4*a13*rhoi*b1*r3 + r1*r3^2)) + b1*d3*r1)/(2*(a13*a31 - a13*r3))];
e6 = [0; 0; rhoi/d3];
E = [e1,e2,e3,e4,e5,e6]; E = double(E)'; E(E>-1e-12 & E<1e-12) = 0;
names = {'eq1';'eq2';'eq3';'eq4';'eq5';'eq6'};
fprintf('%-11s | %-9s | %-9s | %-9s|\n','Equilibria', ' x*', ' y*', ' z*');
fprintf('-------------------------------------------------\n');
for i = 1:length(E)
fprintf('%-10s (%10.3e ,%10.3e ,%10.3e)\n', names{i}, E(i,1), E(i,2), E(i,3));
end
x = E(:,1); y = E(:,2); z = E(:,3); lr = zeros(length(E),3); li =
zeros(length(E),3); stability = cell(length(E),1);
for i = 1:length(E)
J = [- a12*y(i) - a13*z(i) - r1*(b1*x(i) - 1) - b1*r1*x(i), -a12*x(i), -a13*x(i);
- a21*y(i), - a21*x(i) - r2*(b2*y(i) - 1) - b2*r2*y(i), 0;
- z(i)*(a31 - r3), 0, - d3 - x(i)*(a31 - r3)];

lr(i,:) = real(eig(J)); lr(lr>-1e-12 & lr<1e-12) = 0;
li(i,:) = imag(eig(J));
if all(lr(i,:) < 0)
stability{i,1} = 'stable';
elseif any(lr(i,:) > 0)
stability{i,1} = 'unstable';
else
stability{i,1} = 'unidentified';
end
end
lambda = lr + 1i*li;

fprintf('\n\n%-10s | %-16s | %-16s | %-16s| %-10s|\n','Equilibria', ' lambda 1', '
lambda 2', ' lambda 3', 'stability');
```

```
fprintf('-----------------------------------------------------------------------
------\n');
for i = 1:length(E)
fprintf('%-10s %+.1e%+.1ei %+.1e%+.1ei %+.1e%+.1ei %-10s\n', names{i},
real(lambda(i,1)),imag(lambda(i,1)), ...
real(lambda(i,2)),imag(lambda(i,2)), real(lambda(i,3)),imag(lambda(i,3)),
stability{i});
end
end
```